

(52) **U.S. Cl.**

CPC *G06F 12/0802* (2013.01); *G06F 12/1009*
(2013.01); *G06F 2212/45* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,194,736	B2 *	12/2021	Kang	G06F 12/0246
11,693,781	B2	7/2023	Minopoli et al.	
2016/0188490	A1 *	6/2016	Samih	G11C 7/1072
				711/135
2020/0226072	A1	7/2020	Kang et al.	
2022/0004489	A1	1/2022	Cariello	
2022/0058134	A1 *	2/2022	Minopoli	G06F 12/0866
2022/0156185	A1 *	5/2022	Colella	G06F 12/0223
2023/0062372	A1 *	3/2023	Cariello	G06F 3/0659

OTHER PUBLICATIONS

International Search Report and Written Opinion—PCT/US2024/
032850—ISA/EPO—Sep. 2, 2024.

* cited by examiner

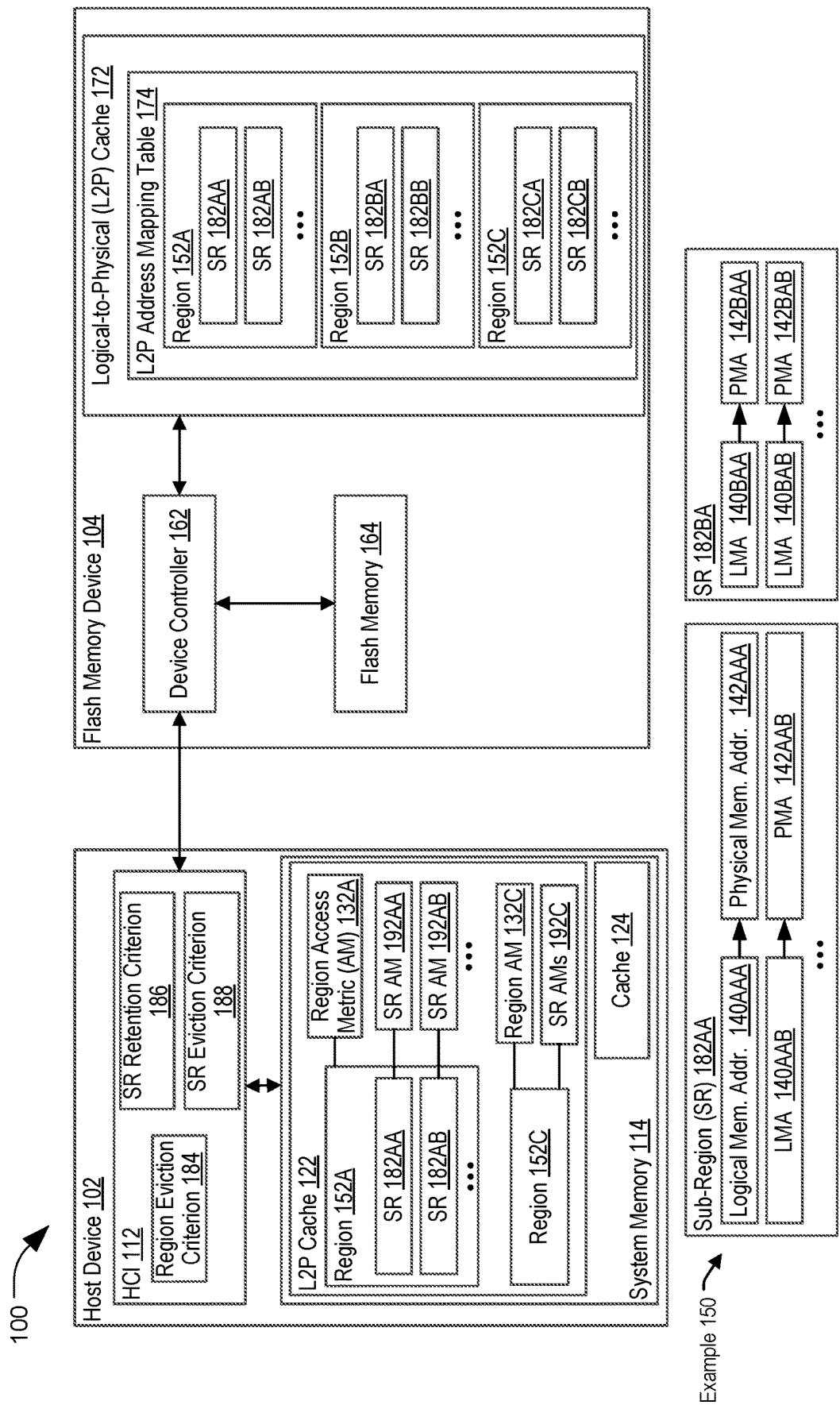


FIG. 1

200 →

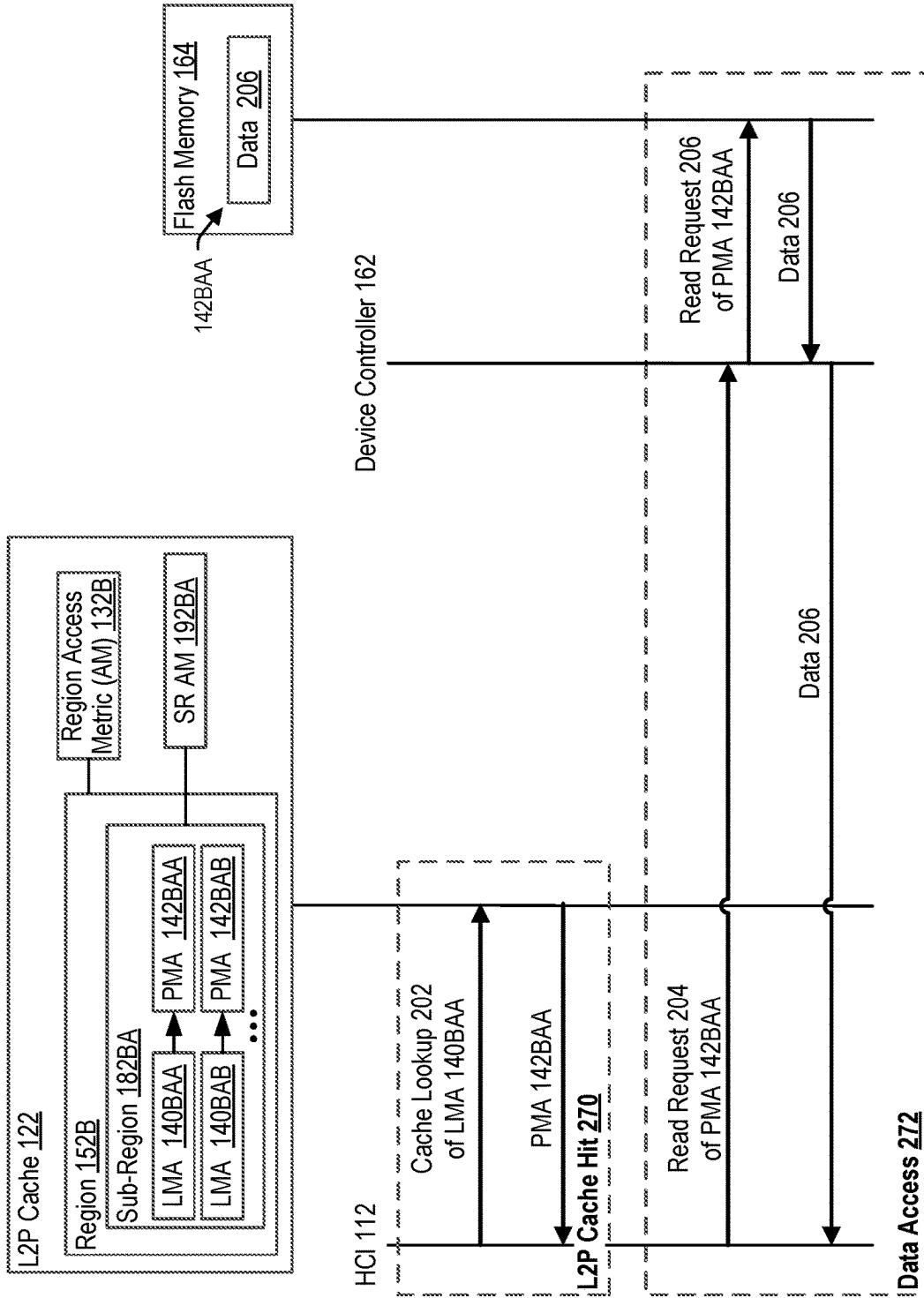


FIG. 2

300 →

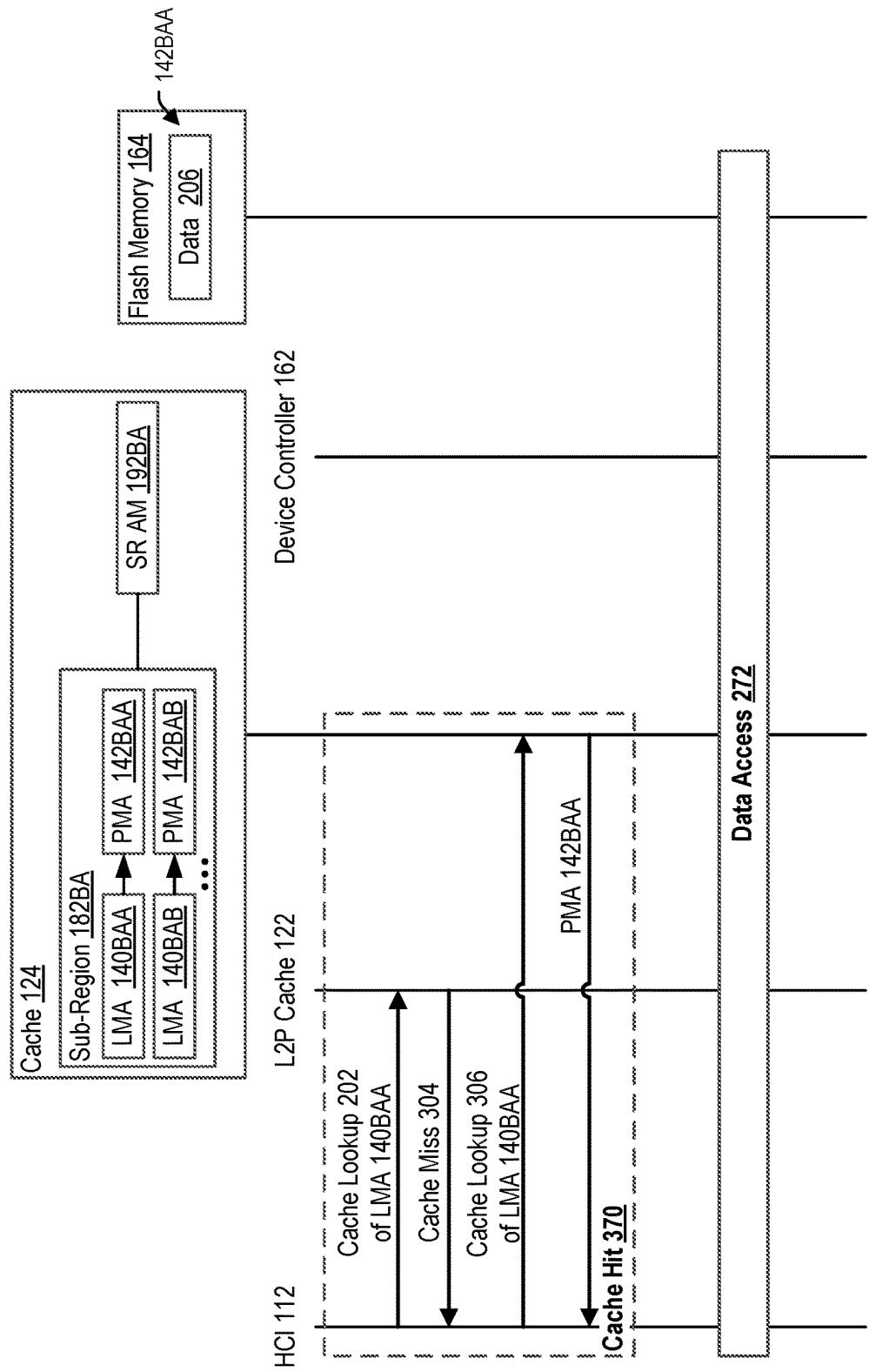


FIG. 3

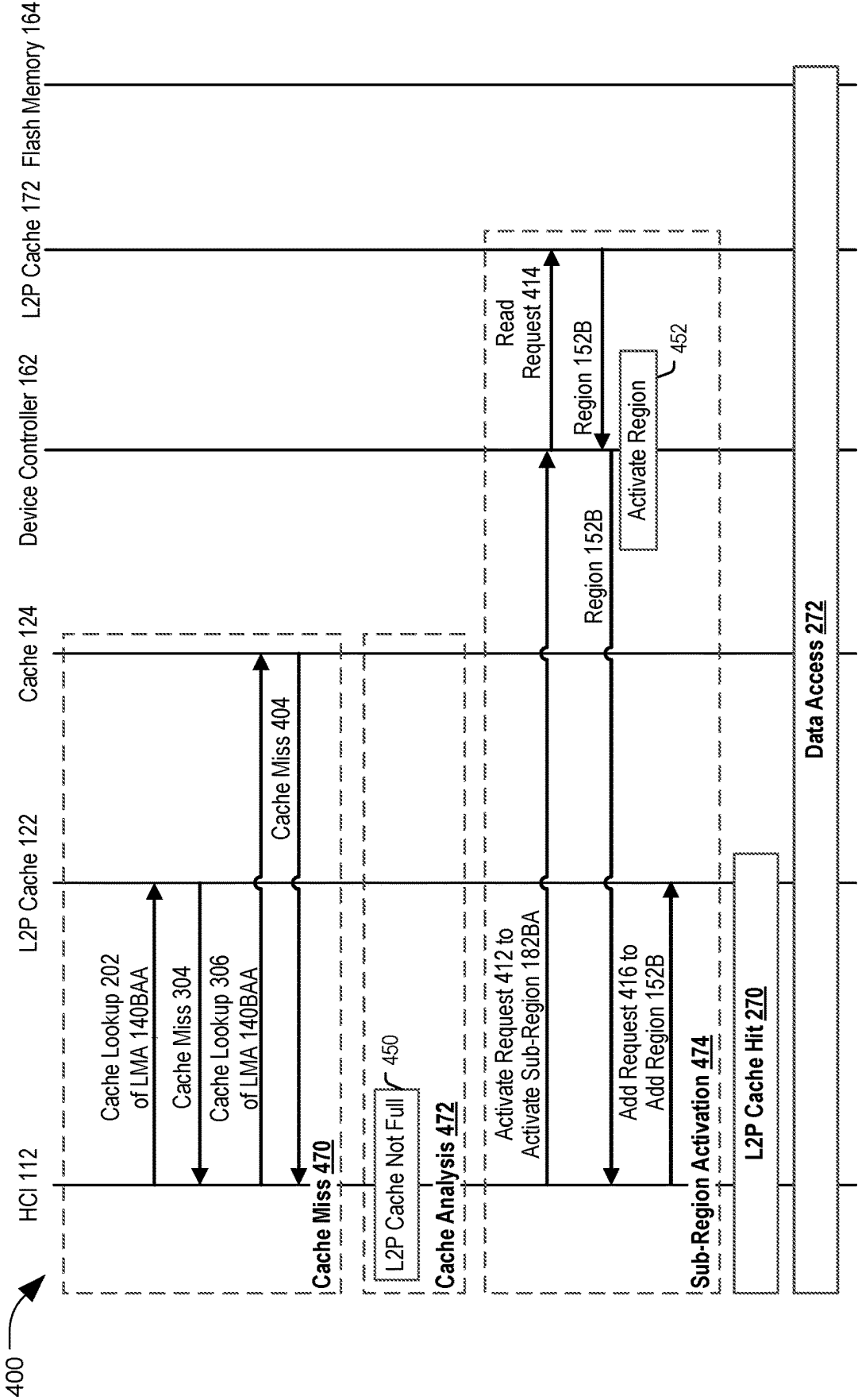
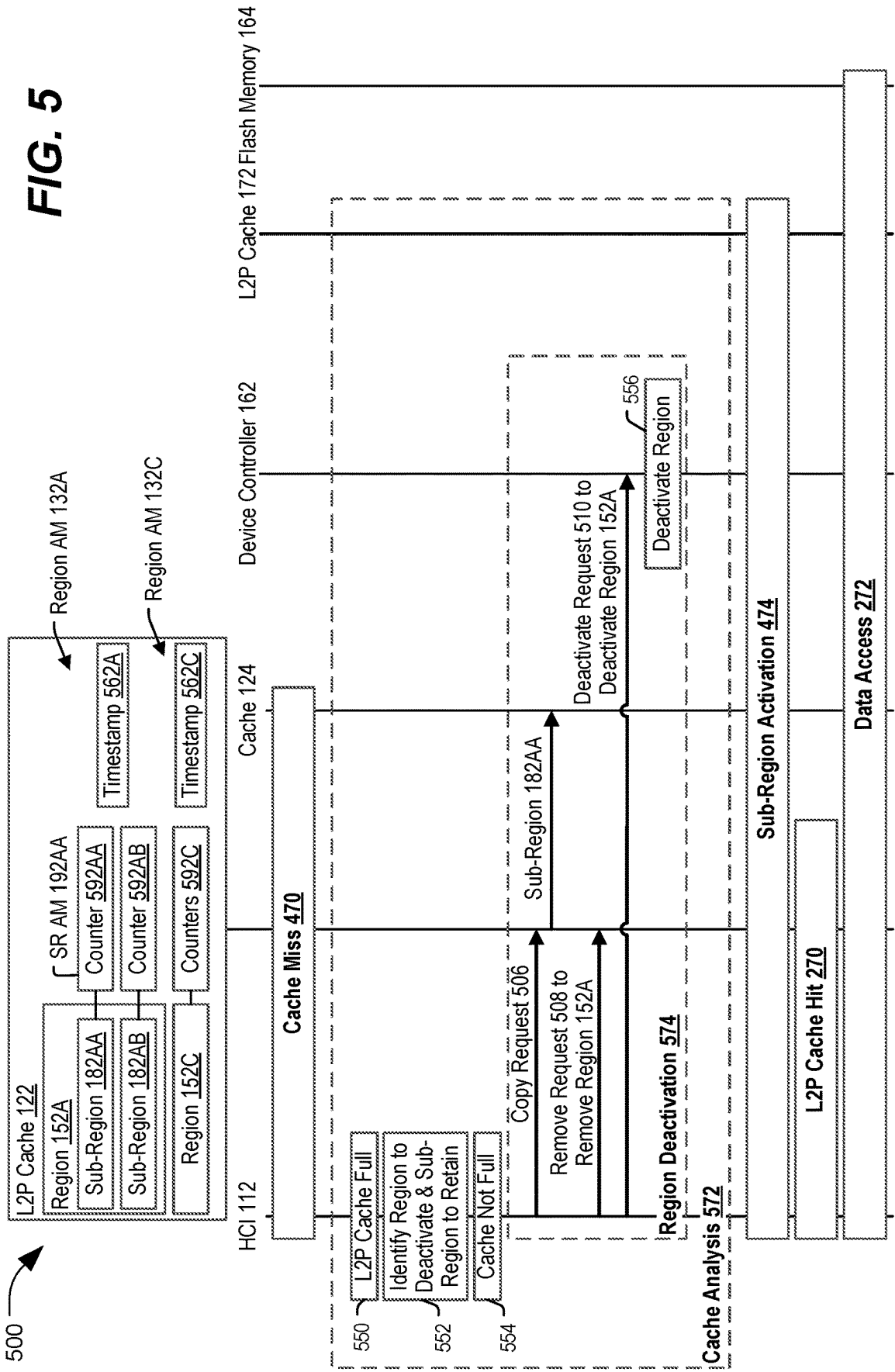
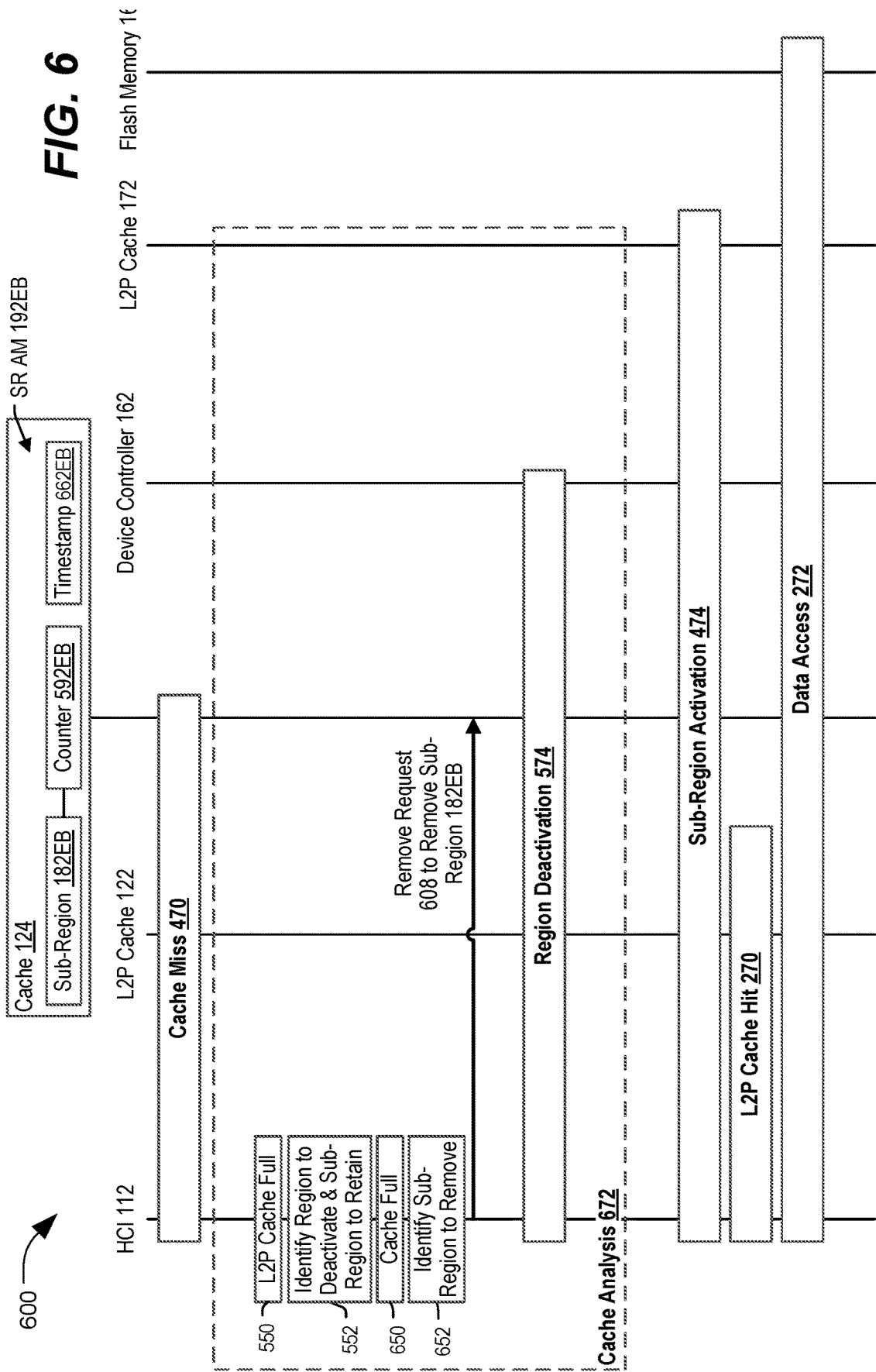


FIG. 4





700 —→

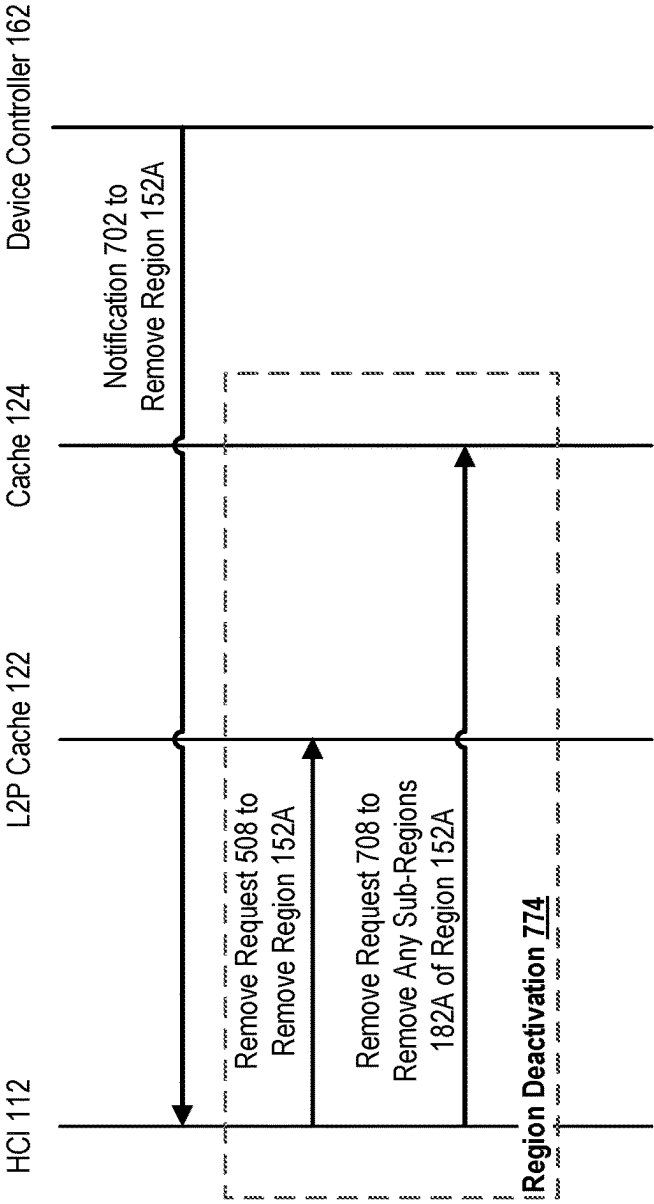


FIG. 7

800 

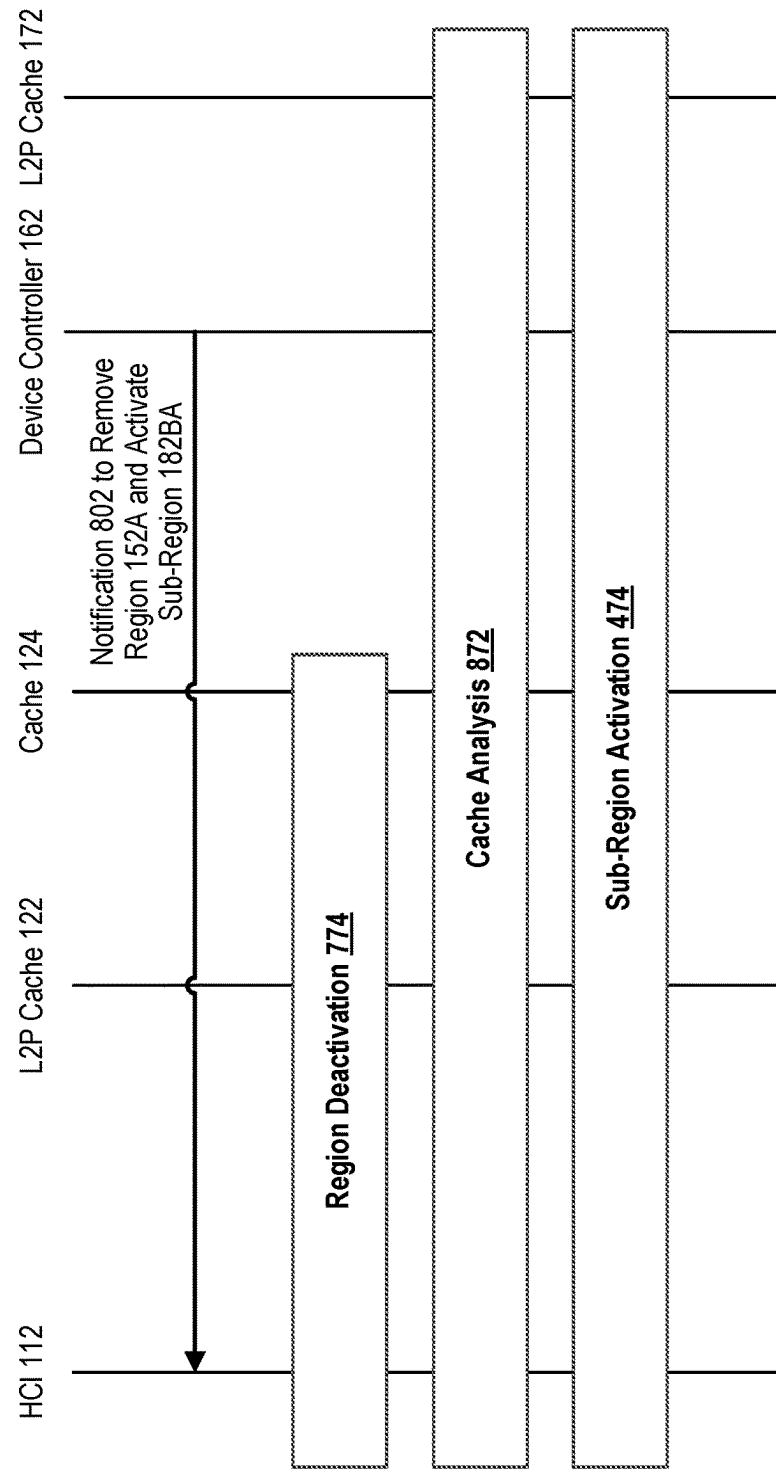


FIG. 8

900 — 

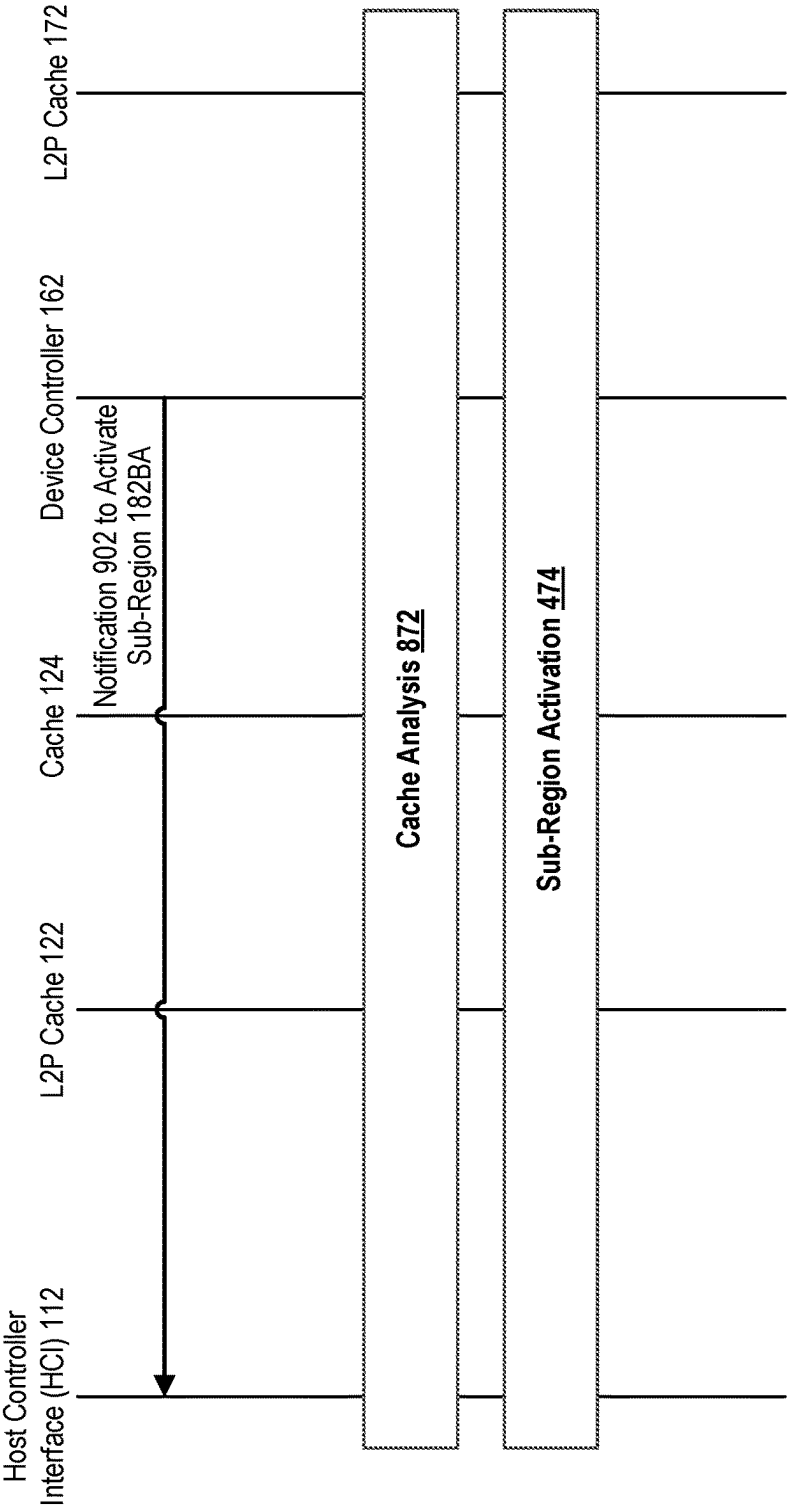


FIG. 9

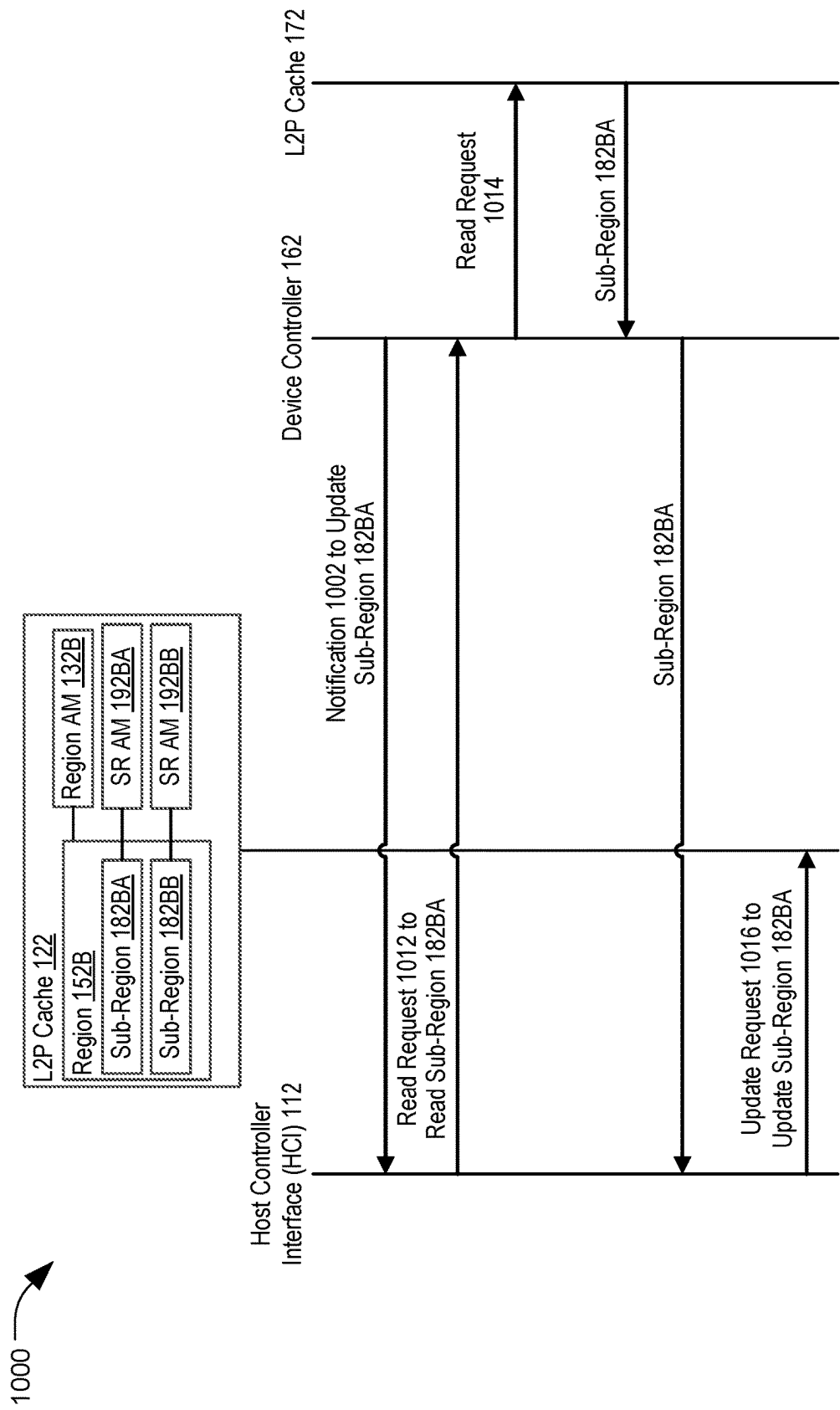


FIG. 10

1100 →

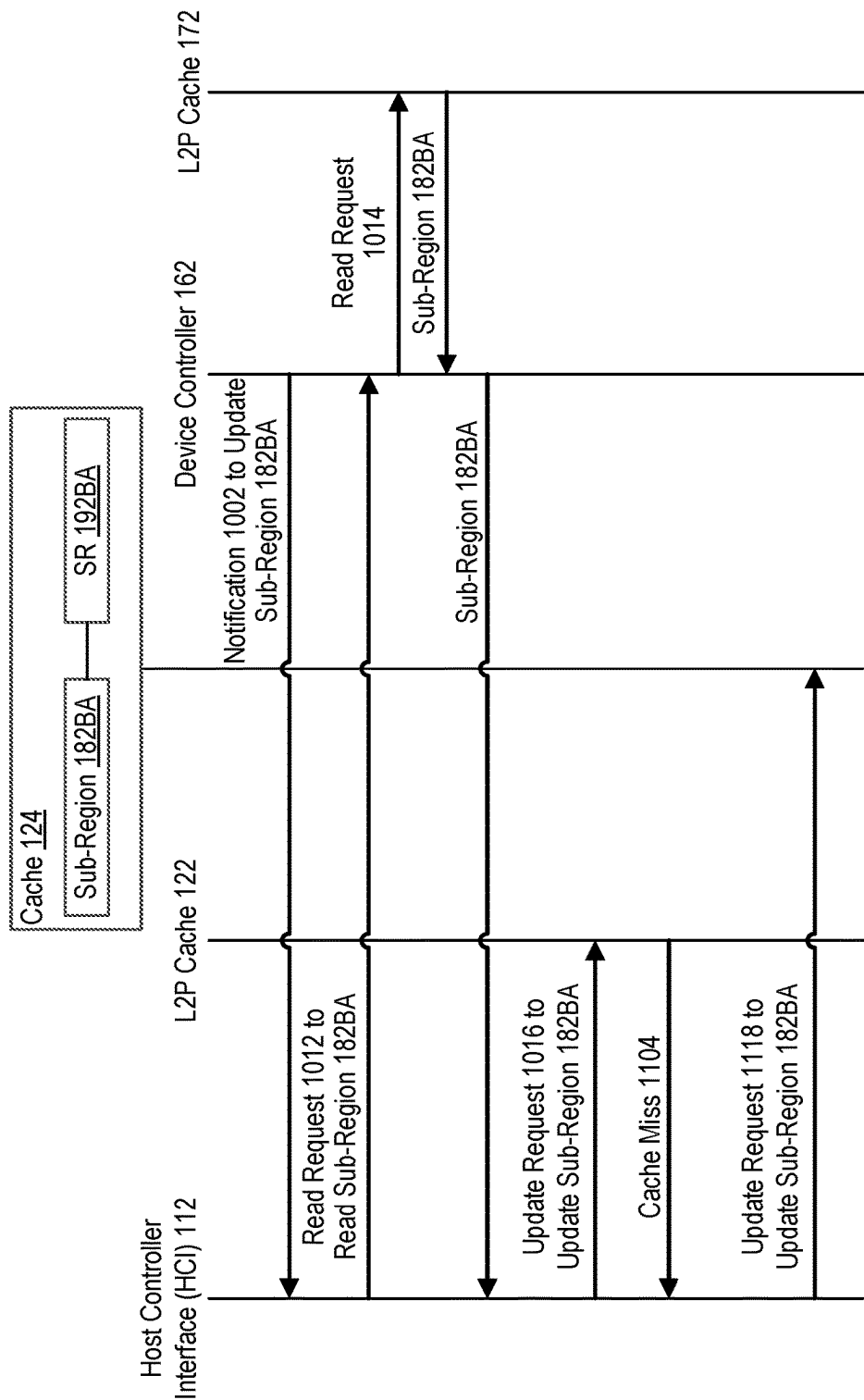
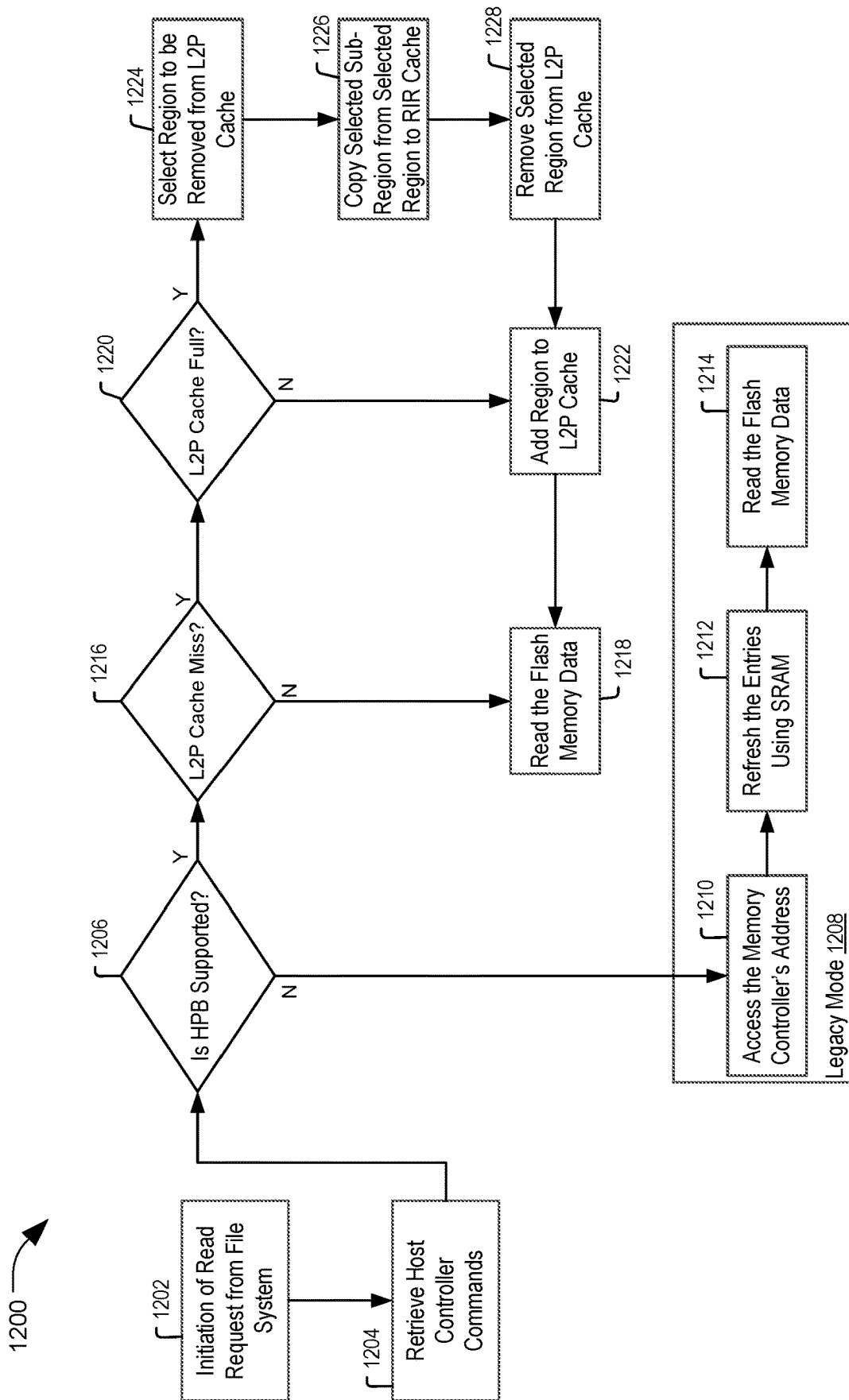
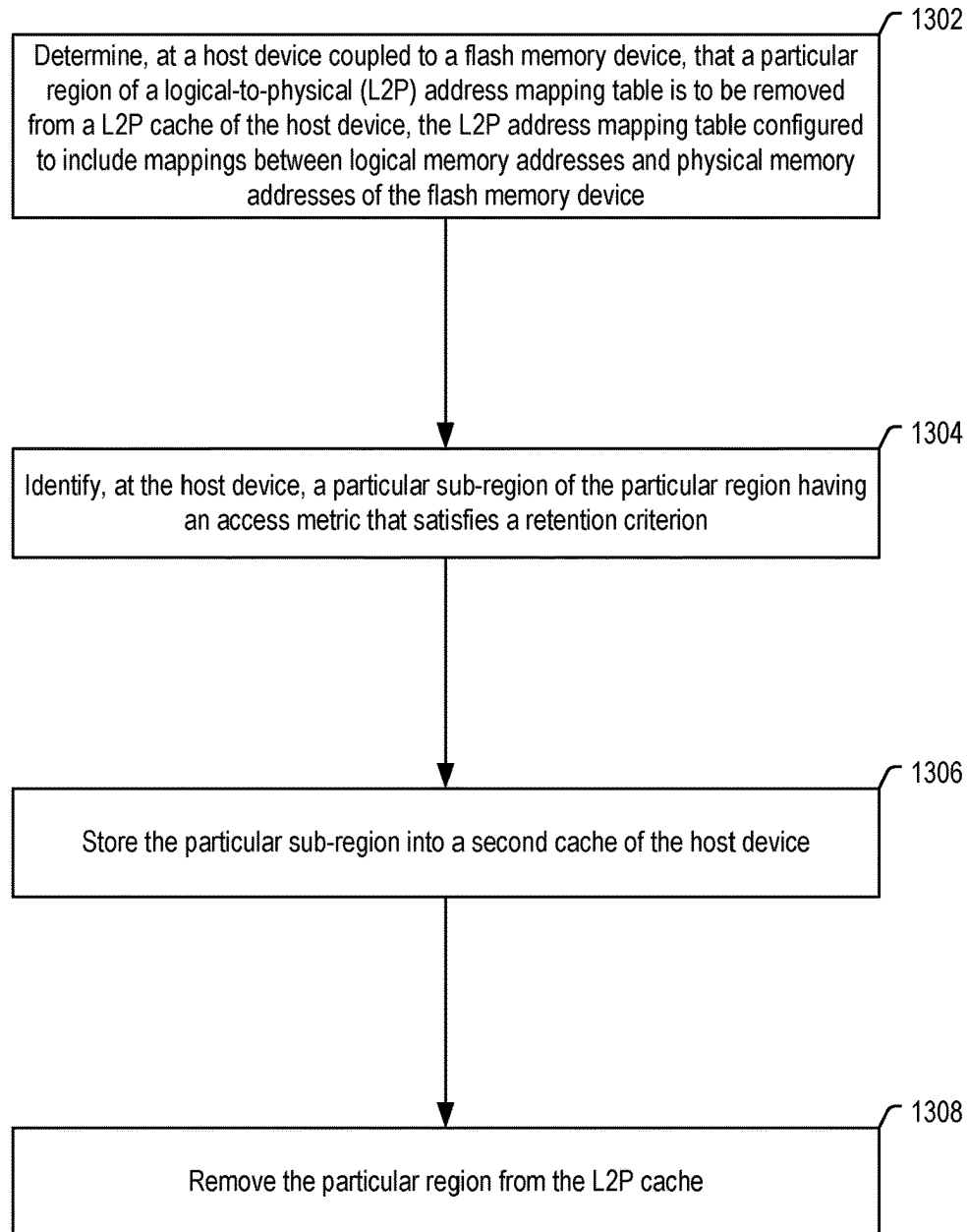


FIG. 11

**FIG. 12**

1300 **FIG. 13**

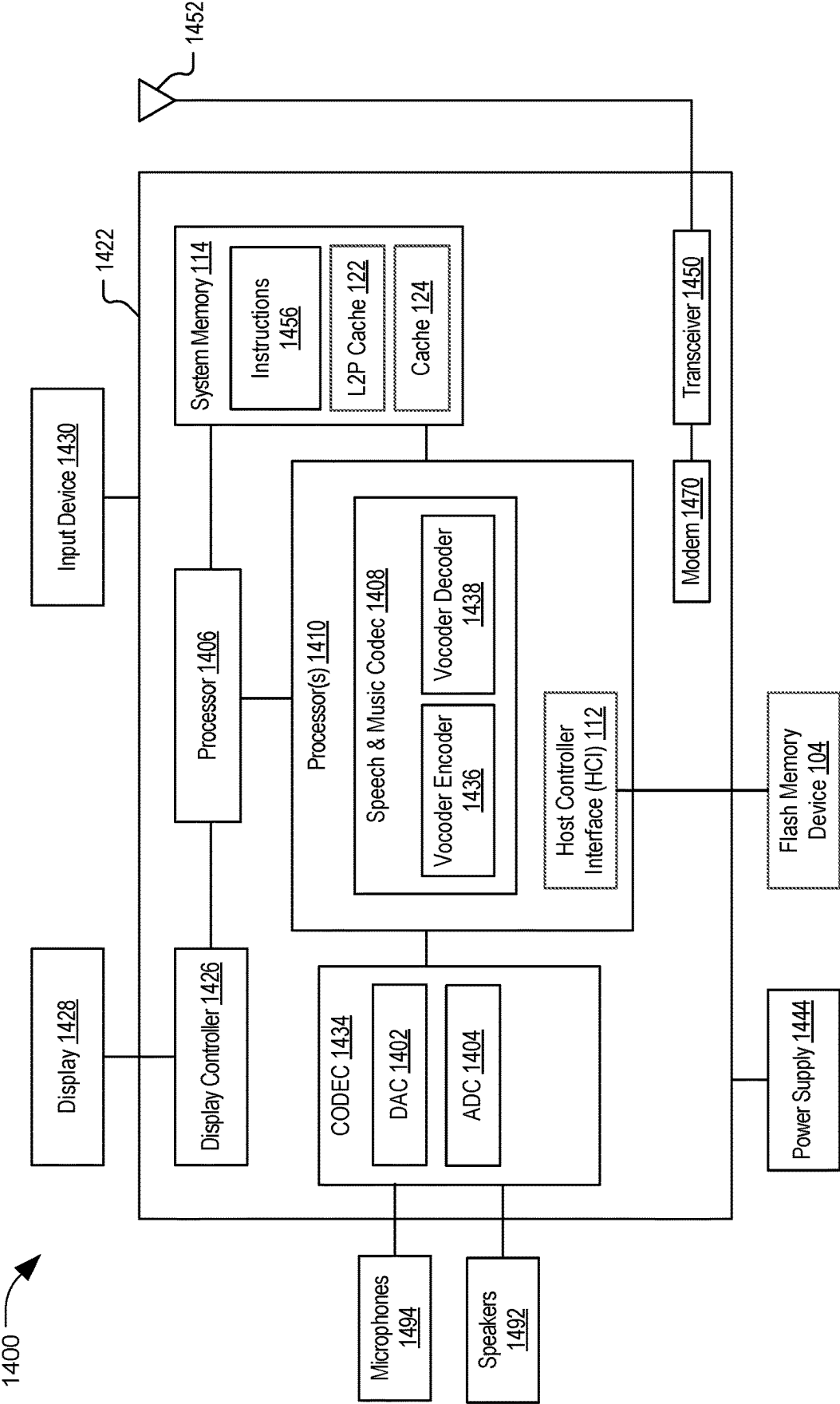


FIG. 14

1

HOST DEVICE CACHING OF FLASH MEMORY ADDRESS MAPPINGS

I. FIELD

The present disclosure is generally related to caching, at a host device, address mappings of a flash memory device.

II. DESCRIPTION OF RELATED ART

Advances in technology have resulted in smaller and more powerful computing devices. For example, there currently exist a variety of portable personal computing devices, including wireless telephones such as mobile and smart phones, tablets and laptop computers that are small, lightweight, and easily carried by users. These devices can communicate voice and data packets over wireless networks. Further, many such devices incorporate additional functionality such as a digital still camera, a digital video camera, a digital recorder, and an audio file player. Also, such devices can process executable instructions, including software applications, such as a web browser application, that can be used to access the Internet. As such, these devices can include significant computing capabilities.

Such computing devices often incorporate functionality to operate as a host device that can store and retrieve data from a flash memory device. For example, the host device can retrieve audio, an image, a video, a document, etc. stored on the flash memory device. Latency associated with reading data from the flash memory device can adversely impact user experience at the host device.

III. SUMMARY

According to one implementation of the present disclosure, a host device includes system memory that includes a logical-to-physical (L2P) cache and a second cache. The host device also includes a host controller interface (HCI) configured to be coupled to a flash memory device. The HCI is configured to determine that a particular region of a L2P address mapping table is to be removed from the L2P cache. The L2P address mapping table is configured to include mappings between logical memory addresses and physical memory addresses of the flash memory device. The HCI is also configured to identify a particular sub-region of the particular region having an access metric that satisfies a retention criterion. The HCI is further configured to store the particular sub-region into the second cache. The HCI is also configured to remove the particular region from the L2P cache.

According to another implementation of the present disclosure, a method includes, at a host device coupled to a flash memory device, determining that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of the host device. The L2P address mapping table is configured to include mappings between logical memory addresses and physical memory addresses of the flash memory device. The method also includes identifying a particular sub-region of the particular region having an access metric that satisfies a retention criterion. The method further includes storing the particular sub-region into a second cache of the host device. The method also includes removing the particular region from the L2P cache.

According to another implementation of the present disclosure, a non-transitory computer-readable medium stores instructions that, when executed by one or more processors,

2

cause the one or more processors to determine that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of a host device. The L2P address mapping table is configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device. The instructions further cause the one or more processors to identify a particular sub-region of the particular region having an access metric that satisfies a retention criterion. The instructions also cause the one or more processors to store the particular sub-region into a second cache of the host device. The instructions further cause the one or more processors to remove the particular region from the L2P cache.

According to another implementation of the present disclosure, an apparatus includes means for determining that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of a host device. The L2P address mapping table is configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device. The apparatus also includes means for identifying a particular sub-region of the particular region having an access metric that satisfies a retention criterion. The apparatus further includes means for storing the particular sub-region into a second cache of the host device. The apparatus also includes means for removing the particular region from the L2P cache.

Other aspects, advantages, and features of the present disclosure will become apparent after review of the entire application, including the following sections: Brief Description of the Drawings, Detailed Description, and the Claims.

IV. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a particular illustrative aspect of a system operable to perform host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 2 is a ladder diagram of an illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 3 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 4 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 5 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 6 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 7 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 8 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 9 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address

3

mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 10 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 11 is a ladder diagram of another illustrative aspect of operations associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

FIG. 12 is a diagram of a particular implementation of a method of host device caching of address mappings of a flash memory device that may be performed by the device of FIG. 1, in accordance with some examples of the present disclosure.

FIG. 13 is a diagram of another particular implementation of a method of host device caching of address mappings of a flash memory device that may be performed by the device of FIG. 1, in accordance with some examples of the present disclosure.

FIG. 14 is a block diagram of a particular illustrative example of a device that is operable to perform host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure.

V. DETAILED DESCRIPTION

Computing devices often incorporate functionality to store and retrieve data from a flash memory device. For example, a host device can retrieve audio, an image, a video, a document, etc. stored on the flash memory device. The flash memory device maps logical memory addresses (e.g., file system addresses) used by the host device to physical memory addresses of the flash memory. Mapping logical memory addresses to physical memory addresses provides advantages in managing flash memory. For example, the flash memory device can move frequently accessed data to different memory cells to balance memory cell usage and prolong overall memory cell life.

The flash memory device includes a logical-to-physical (L2P) cache that stores a L2P address mapping table that includes mappings of the logical memory addresses to corresponding physical memory addresses of the flash memory. To access a logical memory address, the host device sends a memory access request indicating the logical memory address to the flash memory device. The flash memory device uses the L2P address mapping table to obtain a physical memory address corresponding to the logical memory address, and performs the memory access at the physical memory address. To reduce a memory access latency associated with the flash memory device obtaining the physical memory address from the L2P address mapping table for each memory access, the host device can store some of the address mappings at a L2P cache at the host device and send a memory access request with the physical memory address to the flash memory device.

In an example, the logical memory address space is divided into regions, and each region includes a plurality of sub-regions. The host device activates a first sub-region in response to determining that a first logical memory address to be accessed is included in the first sub-region and that the first sub-region is not stored in the L2P cache at the host device. Activating the first sub-region includes sending, to the flash memory device, a first request to activate the first sub-region. The flash memory device, in response to receiving the first request, uses the L2P cache at the flash memory device to determine mappings of a first region that includes

4

the first sub-region and sends the mappings to the host device. The host device stores the mappings of the first region in the L2P cache at the host device. As used herein, an “active sub-region” refers to a sub-region that is stored in the L2P cache at the host device. As used herein, an “active region” refers to a region with at least one sub-region stored in the L2P cache at the host device. A “sub-region” is used interchangeably with “mappings of the sub-region.” A “region” is used interchangeably with “mappings of the region.”

After storing the first region into the L2P cache at the host device, the host device sends, to the flash memory device, a second request to activate a second sub-region. For example, the host device sends the second request in response to determining that a second logical memory address to be accessed is included in the second sub-region and that the second sub-region is not stored in the L2P cache at the host device. The flash memory device, in response to receiving the second request, uses the L2P cache at the flash memory device to determine mappings of a second region that includes the second sub-region and sends the mappings of the second region to the host device.

The host device, in response to receiving the second region and determining that the L2P cache at the host device is full, selects a region to evict from the L2P cache to make space for the second region. Various eviction criteria can be used to select the region to evict. For example, a least accessed region can be selected for eviction, a region that is added earliest to the L2P cache can be selected for eviction, etc. The selected region (e.g., the first region) can include one or more sub-regions that are more frequently accessed at the host device than other sub-regions. Evicting all the sub-regions of the selected region can increase the chance of a cache miss at the host device and increase memory access latency associated with requesting mappings of sub-regions of the evicted region.

Systems and methods of hybrid caching at the host device are disclosed. For example, the host device includes a second cache that is used to store one or more sub-regions of a region that is evicted from the L2P cache at the host device. To illustrate, when the first region is to be evicted to make room for the second region, the host device selects a first sub-region of the first region based on a retention criterion and copies the first sub-region to the second cache prior to replacing the first region with the second region in the L2P cache of the host device.

Subsequently, to perform a memory access of a first logical memory address of the first sub-region, such as to read data from the flash memory device, the host device accesses evicted data from the L2P cache that has been copied into the second cache and reads a first physical memory address associated with the first logical memory address from the second cache. The host device sends, to the flash memory device, a memory access request (e.g., a read instruction) indicating the first physical memory address from which to read the requested data. Because the host device was able to retrieve the first physical address from the second cache even though the first region had been evicted from the L2P cache, the host device was able to avoid the delay associated with requesting the first region from the flash memory device and re-storing the first region into the L2P cache at the host device. In other words, storing the first sub-region in the second cache reduces memory access latency associated with retrieving mappings of the first sub-region.

Particular aspects of the present disclosure are described below with reference to the drawings. In the description,

common features are designated by common reference numbers. As used herein, various terminology is used for the purpose of describing particular implementations only and is not intended to be limiting of implementations. For example, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, some features described herein are singular in some implementations and plural in other implementations. To illustrate, FIG. 14 depicts a device 1400 including one or more processors (“processor(s)” 1410 of FIG. 14), which indicates that in some implementations the device 1400 includes a single processor 1410 and in other implementations the device 1400 includes multiple processors 1410. For ease of reference herein, such features are generally introduced as “one or more” features and are subsequently referred to in the singular or optional plural (as indicated by “(s)”) unless aspects related to multiple of the features are being described.

In some drawings, multiple instances of a particular type of feature are used. Although these features are physically and/or logically distinct, the same reference number is used for each, and the different instances are distinguished by addition of a letter to the reference number. When the features as a group or a type are referred to herein e.g., when no particular one of the features is being referenced, the reference number is used without a distinguishing letter. However, when one particular feature of multiple features of the same type is referred to herein, the reference number is used with the distinguishing letter. For example, referring to FIG. 1, multiple regions are illustrated and associated with reference numbers 152A, 152B, and 152C. When referring to a particular one of these regions, such as a region 152A, the distinguishing letter “A” is used. However, when referring to any arbitrary one of these regions or to these regions as a group, the reference number 152 is used without a distinguishing letter.

As used herein, the terms “comprise,” “comprises,” and “comprising” may be used interchangeably with “include,” “includes,” or “including.” Additionally, the term “wherein” may be used interchangeably with “where.” As used herein, “exemplary” indicates an example, an implementation, and/or an aspect, and should not be construed as limiting or as indicating a preference or a preferred implementation. As used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not by itself indicate any priority or order of the element with respect to another element, but rather merely distinguishes the element from another element having a same name (but for use of the ordinal term). As used herein, the term “set” refers to one or more of a particular element, and the term “plurality” refers to multiple (e.g., two or more) of a particular element.

As used herein, “coupled” may include “communicatively coupled,” “electrically coupled,” or “physically coupled,” and may also (or alternatively) include any combinations thereof. Two devices (or components) may be coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) directly or indirectly via one or more other devices, components, wires, buses, networks (e.g., a wired network, a wireless network, or a combination thereof), etc. Two devices (or components) that are electrically coupled may be included in the same device or in different devices and may be connected via electronics, one or more connectors, or inductive coupling, as illustrative, non-limiting examples. In some implementations, two devices (or components) that are communicatively coupled, such as in electrical communication, may send and receive signals

(e.g., digital signals or analog signals) directly or indirectly, via one or more wires, buses, networks, etc. As used herein, “directly coupled” may include two devices that are coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) without intervening components.

In the present disclosure, terms such as “determining,” “calculating,” “estimating,” “shifting,” “adjusting,” etc. may be used to describe how one or more operations are performed. It should be noted that such terms are not to be construed as limiting and other techniques may be utilized to perform similar operations. Additionally, as referred to herein, “obtaining,” “generating,” “calculating,” “estimating,” “using,” “selecting,” “accessing,” and “determining” may be used interchangeably. For example, “obtaining,” “generating,” “calculating,” “estimating,” or “determining” a parameter (or a signal) may refer to actively generating, estimating, calculating, or determining the parameter (or the signal) or may refer to using, selecting, or accessing the parameter (or signal) that is already generated, such as by another component or device.

Referring to FIG. 1, a particular illustrative aspect is shown of a system 100 that is configured to perform host device caching of address mappings of a flash memory device. The system 100 includes a host device 102 that is configured to be coupled to a flash memory device 104.

The flash memory device 104 includes a device controller 162 that is coupled to a flash memory 164 (e.g., NAND memory). The flash memory device 104 includes a L2P cache 172 (e.g., static random access memory (SRAM)). The L2P cache 172 is configured to store a L2P address mapping table 174. The L2P address mapping table 174 is configured to include mappings between logical memory addresses 140 (e.g., file system addresses at the host device 102) and physical memory addresses 142 of the flash memory 164.

In an example, a logical memory address space is logically divided into regions 152, such as a region 152A, a region 152B, a region 152C, one or more additional regions, or a combination thereof. Each region 152 includes one or more sub-regions (SRs) 182. For example, the region 152A includes sub-regions 182A, such as a sub-region (SR) 182AA, a sub-region 182AB, one or more additional sub-regions, or a combination thereof. As another example, the region 152B includes sub-regions 182B, such as a sub-region 182BA, a sub-region 182BB, one or more additional sub-regions, or a combination thereof. As yet another example, the region 152C includes sub-regions 182C, such as a sub-region 182CA, a sub-region 182CB, one or more additional sub-regions, or a combination thereof.

In a particular implementation, each of the regions 152 includes the same (e.g., a pre-determined) count of sub-regions 182. According to other implementations, the region 152A includes a first count of sub-regions 182 that is distinct from a second count of sub-regions 182 included in the region 152B. In an example, the sub-region 182AA corresponds to a first logical memory address range and the sub-region 182BA corresponds to a second logical memory address range that is distinct from the first logical memory address range. In a particular implementation, a first size of the first logical memory address range is equal to a second size of the second logical memory address range. In another implementation, the first size of the first logical memory address range is greater than the second size of the second logical memory address range.

According to some implementations, a size of the L2P cache 172 is greater than a size of the L2P cache 122. In a particular aspect, the size of the L2P cache 122 is based on

user input, default data, a configuration setting, or a combination thereof. In a particular example, if each of the regions 152 has a predetermined region size (e.g., bHPBRegionSize=2 gigabytes (GB)) and with a pre-determined logical block size (e.g., bHPBSubRegion Size=4 kilobyte (KB)), each of the regions 152 includes a pre-determined count of sub-regions 182 (e.g., 2 GB/4 KB=500000). With up to a threshold count of active regions (e.g., wDeviceMaxActiveHPBRegions=10) supported and a pre-determined size (e.g., 8 bytes) of each entry in the L2P cache 122, the host device 102 can allocate a first size (e.g., 500000*10*8 bytes=40 MB) to the L2P cache 122. If the each of the regions 152 has a smaller predetermined region size (e.g., bHPBRegionSize=32 megabytes (MB)), each of the regions 152 includes a smaller pre-determined count of sub-regions 182 (e.g., 32 MB/4 KB=8000) and the same size (e.g., 8000*640*8 bytes=40 MB) of the L2P cache 122 can be used to store a greater count (e.g., 640) of active regions.

Each of the sub-regions 182 includes one or more mappings between the logical memory addresses 140 and the physical memory addresses 142. In an example 150, the sub-region 182AA includes mappings between logical memory addresses 140AA and physical memory addresses 142AA, and the sub-region 182BA includes mappings between logical memory addresses 140BA and physical memory addresses 142BA. To illustrate, the sub-region 182AA includes a mapping between a logical memory address (LMA) 140AAA and a physical memory address (PMA) 142AAA, a mapping between a logical memory address 140AAB and a physical memory address 142AAB, one or more additional mappings, or a combination thereof. The sub-region 182BA includes a mapping between a logical memory address 140BAA and a physical memory address 142BAA, a mapping between a logical memory address 140BAB and a physical memory address 142BAB, one or more additional mappings, or a combination thereof.

The host device 102 includes a host controller interface (HCI) 112 that is coupled to a system memory 114 (e.g., a system-on-chip (SOC) dynamic random access memory (DRAM)). The system memory 114 includes a logical-to-physical (L2P) cache 122 and a cache 124. In a particular implementation, the L2P cache 122 corresponds to a Host Performance Booster (HPB) cache. In a particular implementation, the cache 124 is implemented in a recently inactivated region (RIR) partition of the system memory 114. The HCI 112 is configured to be coupled to the flash memory device 104. In an example, the HCI 112 is configured to communicate with the device controller 162 of the flash memory device 104. According to some implementations, the HCI 112 is configured to send one or more commands to the device controller 162, and the device controller 162 is configured to send a response to a command received from the HCI 112.

The HCI 112 is configured to use hybrid caching to perform memory accesses (e.g., read or write) of logical memory addresses 140. For example, the HCI 112 is configured to perform cache lookups for mappings of logical memory addresses 140 in the L2P cache 122, perform cache lookups for mappings of logical memory addresses 140 in the cache 124 for cache misses in the L2P cache 122, and retrieve mappings from the L2P cache 122 of the flash memory device 104 for cache misses in the L2P cache 122, as further described with reference to FIGS. 2-6. The device controller 162 is configured to provide mappings to the HCI 112, as further described with reference to FIGS. 4-6.

The HCI 112 is configured to evict a region 152 from the L2P cache 122 based on a region eviction criterion 184, as

further described with reference to FIG. 5. For example, the HCI 112 is configured to, in response to determining that the region 152B is to be stored in the L2P cache 122 and that the L2P cache 122 is full, select the region 152A for eviction from the L2P cache 122 based on the region eviction criterion 184 and select a sub-region 182AA of the region 152A for retention in the cache 124 based on a sub-region retention criterion 186, as further described with reference to FIG. 5. It should be understood that a single sub-region 182 (of the region 152A to be evicted) satisfying the sub-region retention criterion 186 and being retained in the cache 124 is provided as an illustrative example. In other examples, multiple sub-regions 182 (of the region 152A to be evicted) can satisfy the sub-region retention criterion 186 and be retained in the cache 124. In yet other examples, none of the sub-regions 182 (of the region 152A to be evicted) satisfy the sub-region retention criterion 186 and are therefore not retained in the cache 124 when the region 152A is evicted from the L2P cache 122. To illustrate, retention in the cache 124 of a sub-region 182 of the region 152A to be evicted can be selective based on the sub-region retention criterion 186.

The HCI 112 is configured to evict a sub-region 182 from the cache 124 based on a sub-region eviction criterion 188, as further described with reference to FIG. 6. For example, the HCI 112 is configured to, in response to determining that a sub-region 182AA is to be stored in the cache 124 and that the cache 124 is full, select a sub-region 182 for eviction from the cache 124 based on the sub-region eviction criterion 188, as further described with reference to FIG. 6. In a particular aspect, the region eviction criterion 184, the sub-region retention criterion 186, the sub-region eviction criterion 188, or a combination thereof, are based on a user input, a configuration setting, default data, or a combination thereof.

The HCI 112 is configured to, in response to a cache hit resulting from a cache lookup in the L2P cache 122 or a cache lookup in the cache 124, send a memory access request to the device controller 162. The device controller 162 is configured to, in response to receiving the memory access request, perform the memory access (read or write) at the flash memory 164, as further described with reference to FIGS. 2-3.

According to some implementations, the device controller 162 is configured to notify the HCI 112 that a region 152 is to be inactivated, a sub-region 182 is to be activated, or both, as further described with reference to FIGS. 7-9. According to some implementations, the device controller 162 is configured to notify the HCI 112 that a sub-region 182 is to be updated, as further described with reference to FIGS. 10-11.

During operation, in an illustrative example in which the regions 152A and 152B have not yet been stored in the L2P cache 122, the HCI 112 determines that a memory access (e.g., a read or a write) is to be performed of the logical memory address 140AAA (e.g., a logical memory address that is included in the sub-region 182AA of the region 152A). For example, the HCI 112 determines that the memory access is to be performed responsive to receiving a memory access request from a file system of the host device 102.

The HCI 112, in response to determining that a memory access is to be performed of the logical memory address 140AAA, performs a cache lookup of the logical memory address 140AAA in the L2P cache 122. Performing a cache lookup at the L2P cache 122 includes comparing the logical memory address 140AAA of data to be accessed to logical addresses (e.g., logical address ranges) of the regions 152 stored in the L2P cache 122. The HCI 112, in response to

determining that the logical memory address **140AAA** does not match (e.g., is not included in) the logical addresses (e.g., the logical address ranges) of any of the regions **152** stored in the L2P cache **122**, detects a cache miss resulting from the cache lookup in the L2P cache **122**.

The HCI **112**, in response to detecting the cache miss resulting from the cache lookup in the L2P cache **122**, performs a cache lookup of the logical memory address **140AAA** in the cache **124**. Performing a cache lookup at the cache **124** includes comparing the logical memory address **140AAA** to logical addresses (e.g., logical address ranges) of the sub-regions **182** stored in the cache **124**. The HCI **112**, in response to determining that the logical memory address **140AAA** does not match (e.g., is not included in) the logical addresses (e.g., the logical address ranges) of any of the sub-regions **182** stored in the cache **124**, detects a cache miss resulting from the cache lookup in the cache **124**.

The HCI **112**, in response to the cache miss resulting from the cache lookup in the cache **124**, performs a sub-region activation of the sub-region **182AA** that includes the logical memory address **140AAA**. For example, performing the sub-region activation includes sending an activate request to the device controller **162** to activate the sub-region **182AA**. To illustrate, the HCI **112** sends the activate request subsequent to determining that the L2P cache **122** is not full (e.g., the L2P cache **122** has available space to add the region **152A**).

The device controller **162**, in response to receiving the activate request, retrieves the region **152A** (that includes the sub-region **182AA** to be activated) from the L2P address mapping table **174** and sends the region **152A** to the HCI **112**. The HCI **112** stores the region **152A** in the L2P cache **122** and performs a cache lookup of the logical memory address **140AAA** in the L2P cache **122**.

The HCI **112**, in response to determining that the logical memory address **140AAA** now matches (e.g., is included in) the logical addresses (e.g., logical address ranges) of the region **152A** stored in the L2P cache **122**, detects a cache hit resulting from the cache lookup in the L2P cache **122**. The HCI **112**, in response to detecting the cache hit resulting from the cache lookup in the L2P cache **122**, performs a data access using the physical memory address **142AAA** read from the L2P cache **122**. For example, performing the data access includes reading the physical memory address **142AAA** associated with the logical memory address **140AAA** from the L2P cache **122** and sending a memory access request that includes the physical memory address **142AAA** to the device controller **162**.

The device controller **162**, in response to receiving the memory access request, performs the memory access (e.g., read or write) at the physical memory address **142AAA**. For example, the device controller **162**, in response to receiving a read request, reads data at the physical memory address **142AAA** and provides the data to the HCI **112**. As another example, the device controller **162**, in response to receiving a write request, writes data to the physical memory address **142AAA**. In a particular aspect, the device controller **162** provides a status (e.g., success or failure) of the memory access request to the HCI **112**. In a particular aspect, the HCI **112** provides the data, status, or both, to a component (e.g., the file system) of the host device **102**, another device, or both.

According to some implementations, the HCI **112** maintains sub-region access metrics (AMs) **192** of each of the sub-regions **182** and region access metrics **132** of each of the regions **152** stored in the L2P cache **122**. According to some implementations, the L2P cache **122** corresponds to a HPB

cache and the sub-region access metrics **192** are maintained in a descriptor column of the HPB cache, and the sub-region retention criterion **186** is maintained as an RIR frequency threshold in a control register (e.g., an 8-bit control register).

According to some implementations, the sub-region access metrics **192**, the region access metrics **132**, or both, are associated with a particular time period. For example, the sub-region access metrics **192**, the region access metrics **132**, or both are associated with a sliding time window. In a particular aspect, the HCI **112** resets the sub-region access metrics **192**, the region access metrics **132**, or a combination thereof, based on detecting a reset condition, such as expiration of a timer, a reset of the L2P cache **122**, receipt of a reset command from the host device **102**, or a combination thereof.

In an example, the HCI **112** maintains a sub-region access metric **192AA** of the sub-region **182AA**, a sub-region access metric **192AB** of the sub-region **182AB**, and one or more additional access metrics of one or more additional sub-regions **182A** of the region **152A**. The HCI **112** maintains a region access metric **132A** of the region **152A**. In a particular implementation, the region access metric **132A** is based at least in part on the sub-region access metrics **192A** of the sub-regions **182A** of the region **152A**. As another example, the HCI **112** maintains sub-region access metrics **192C** of the sub-regions **182C** of the region **152C**, and a region access metric **132C** of the region **152C**. In an example, the L2P cache **122** stores the sub-region access metrics **192** of the sub-regions **182** stored in the L2P cache **122** and the region access metrics **132** of the regions **152** stored in the L2P cache **122**. The HCI **112**, in response to performing a memory access of the logical memory address **140AAA**, updates the sub-region access metric **192AA**, the region access metric **132A**, or both.

Continuing the previous example, after storing the region **152A** into the L2P cache **122**, which results in the L2P cache **122** becoming full, the HCI **112** determines that a memory access (e.g., read or write) is to be performed of the logical memory address **140BAA** and performs a cache lookup of the logical memory address **140BAA** in the L2P cache **122**. The HCI **112**, in response to detecting a cache miss resulting from the cache lookup in the L2P cache **122** (because the region **152B** is not in the L2P cache **122**), performs a cache lookup of the logical memory address **140BAA** in the cache **124**. The HCI **112**, in response to detecting a cache miss resulting from the cache lookup in the cache **124**, performs a sub-region activation of the sub-region **182BA**, as further described with reference to FIG. 4. For example, performing the sub-region activation includes sending an activate request to the device controller **162** to activate the sub-region **182BA** that includes the logical memory address **140BAA**.

In a particular aspect, the HCI **112** sends the activate request subsequent to making space in the L2P cache **122** in response to determining that the L2P cache **122** is full. For example, making space in the L2P cache **122** includes selecting a region **152** to be removed from the L2P cache **122**, selecting a sub-region **182** of the selected region **152** to be retained in the cache **124**, copying the selected sub-region **182** to the cache **124**, and removing the selected region **152** from the L2P cache **122**. To illustrate, the HCI **112** determines that the region **152A** is to be removed from the L2P cache **122** based on determining that the region access metric **132A** satisfies the region eviction criterion **184**, as further described with reference to FIG. 4. The HCI **112** identifies the sub-region **182AA** having the sub-region

11

access metric **192AA** that satisfies the sub-region retention criterion **186**, as further described with reference to FIG. 4.

The HCI **112** stores the sub-region **182AA** into the cache **124**, and removes the region **152A** from the L2P cache **122**. In a particular aspect, the HCI **112**, in response to determining that the cache **124** is full, selects a sub-region **182** to be removed from the cache **124** and removes the selected sub-region **182** from the cache **124** prior to storing the sub-region **182AA** to the cache **124**, as further described with reference to FIG. 6. According to some implementations, the HCI **112** maintains sub-region access metrics **192** of each of the sub-regions **182** stored in the cache **124**. In a particular aspect, the cache **124** stores the sub-region access metrics **192** of the sub-regions **182** stored in the cache **124**.

In an example, the HCI **112** selects a sub-region **182** to be removed from the cache **124** based on determining that a sub-region access metric **192** of the sub-region **182** satisfies the sub-region eviction criterion **188**, as further described with reference to FIG. 6. The HCI **112** copies the sub-region **182AA** to the cache **124** after removing the selected sub-region **182** from the cache **124**. In a particular implementation, the HCI **112**, subsequent to adding the sub-region **182AA** to the cache **124**, updates the sub-region access metric **192AA** stored in the cache **124**.

The device controller **162**, in response to the activate request, retrieves the region **152B** (that includes the sub-region **182BA** to be activated) from the L2P address mapping table **174** and sends the region **152B** to the HCI **112**. The HCI **112** stores the region **152B** in the L2P cache **122**, as further described with reference to FIG. 6. To illustrate, the region **152B** (e.g., a replacement region) is stored in the L2P cache **122** after removing the region **152A** from the L2P cache **122**. In a particular aspect, the HCI **112** updates a region access metric **132** of the region **152B** subsequent to adding the region **152B** to the L2P cache **122**. The HCI **112**, after storing the region **152B** in the L2P cache **122**, performs a cache lookup of the logical memory address **140BAA** in the L2P cache **122**.

The HCI **112**, in response to a cache hit resulting from the cache lookup in the L2P cache **122**, reads a physical memory address **142BAA** associated with the logical memory address **140BAA** from the L2P cache **122** and performs a data access using the physical memory address **142BAA**. For example, performing the data access includes sending a memory access request that includes the physical memory address **142BAA** to the device controller **162**. The device controller **162**, in response to receiving the memory access request, performs the memory access (e.g., read or write) at the physical memory address **142BAA**. According to some implementations, the HCI **112**, in response to performing the memory access of the logical memory address **140BAA**, updates a sub-region access metric **192** of the sub-region **182BA**, a region access metric **132** of the region **152B**, or both, stored at the L2P cache **122**.

Continuing the previous example, after evicting the region **152A** and storing the region **152B** into the L2P cache **122** (which is full), along with storing the sub-region **182AA** to the cache **124**, the HCI **112** determines that a memory access (e.g., read or write) is to be performed of the logical memory address **140AAB** and performs a cache lookup of the logical memory address **140AAB** in the L2P cache **122**. The HCI **112**, in response to detecting a cache miss resulting from the cache lookup in the L2P cache **122**, performs a cache lookup of the logical memory address **140AAB** in the cache **124**. The HCI **112**, in response to detecting a match of the logical memory address **140AAB** to the logical addresses (e.g., a logical address range) of the sub-region

12

182AA stored in the cache **124**, detects a cache hit resulting from the cache lookup in the cache **124**.

The HCI **112**, in response to the cache hit resulting from the cache lookup in the cache **124**, reads the physical memory address **142AAB** associated with the logical memory address **140AAB** from the cache **124** and performs a data access using the logical memory address **140AAB**. For example, performing the data access includes sending the physical memory address **142AAB** to the device controller **162** in conjunction with a memory access request to access the data stored at the physical memory address **142AAB** of the flash memory **164**. The device controller **162**, in response to receiving the memory access request, performs the memory access (e.g., read or write) at the physical memory address **142AAB**. According to some implementations, the HCI **112**, in response to performing the memory access of the logical memory address **140AAB**, updates the sub-region access metric **192AA** of the sub-region **182AA** stored in the cache **124**.

According to some implementations, the device controller **162** is configured to notify the HCI **112** that a region **152** is to be inactivated, a sub-region **182** is to be activated, or both, as further described with reference to FIGS. 7-9. The HCI **112**, in response to a notification from the HCI **112**, determines that the region **152** is to be removed, that the sub-region **182** is to be activated, or both. According to some implementations, the device controller **162** is configured to notify the HCI **112** that a sub-region **182** is to be updated, as further described with reference to FIGS. 10-11. The HCI **112**, in response to a notification from the HCI **112**, determines that the sub-region **182** is to be updated.

The sub-region **182AA**, which was copied to the cache **124** in the above example, remains available at the host device **102** in the cache **124** subsequent to the eviction of the region **152A** from the L2P cache **122**. The system **100** thus reduces memory access latency associated with accessing memory address mappings of logical memory addresses included in the sub-region **182AA** subsequent to eviction of the region **152A** from the L2P cache **122**.

It should be understood that the particular order of operations in the above example(s), or in other examples described herein, is provided for purposes of illustration and not of limitation. In other examples, two or more operations can be performed in another order. To illustrate, the HCI **112** evicting the region **152A** prior to sending the activation request to activate the sub-region **182BA** is provided as an illustrative example; in other examples, the HCI **112** can send the activation request prior to evicting the region **152A**.

FIG. 2 is a ladder diagram of an illustrative aspect of operations **200** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **200** are performed by the HCI **112**, the L2P cache **122**, the host device **102**, the device controller **162**, the flash memory **164**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

In the example illustrated in FIG. 2, a memory access is to be performed of the logical memory address **140BAA** included in the sub-region **182BA** of the region **152B**, and the region **152B** is stored in the L2P cache **122**. The operations **200** include a L2P cache hit **270** followed by a data access **272**. The L2P cache hit **270** includes the HCI **112** performing a cache lookup **202** of the logical memory address **140BAA** in the L2P cache **122**. The HCI **112** detects a cache hit in response to determining that the logical memory address **140BAA** matches (e.g., is included in) the

13

logical memory addresses **140BA** of the sub-region **182BA** of the region **152B** stored in the L2P cache **122**. The HCI **112**, in response to detecting the cache hit, reads the physical memory address **142BAA** associated with the logical memory address **140BAA** from the L2P cache **122**.

The data access **272** includes the HCI **112** performing a memory access at the physical memory address **142BAA** of the flash memory **164**. For example, the HCI **112** sends the physical memory address **142BAA** in conjunction with a read request **204** to the device controller **162**. The device controller **162**, in response to receiving the read request **204** indicating the physical memory address **142BAA**, performs a read of data **206** from the physical memory address **142BAA** at the flash memory **164**. The device controller **162** provides the data **206** to the HCI **112**.

In a particular implementation, the HCI **112**, in response to performing the memory access, updates a sub-region access metric **192BA** of the sub-region **182BA**, a region access metric **132B** of the region **152B**, or both. In a particular implementation, the sub-region access metric **192BA** indicates a count of memory accesses to the sub-region **182BA** during a particular time period and the HCI **112** updates (e.g., increments by 1) the count of memory accesses to the sub-region **182BA**. In a particular implementation, the region access metric **132B** indicates a count of memory accesses to the region **152B** during a particular time period and the HCI **112** updates (e.g., increments by 1) the count of memory accesses to the region **152B**.

In a particular implementation, the sub-region access metric **192BA** indicates a time of a most recent memory access of the sub-region **182BA**, and the HCI **112** updates the sub-region access metric **192BA** at a first time to indicate a sub-region memory access time associated with the most recent memory access to the sub-region **182BA**. In some aspects, the sub-region memory access time includes the first time, a memory access time received from the device controller **162**, a receipt time of a notification from the device controller **162**, or a combination thereof. In a particular implementation, the region access metric **132B** indicates a time of a most recent memory access of the region **152B**, and the HCI **112** updates the region access metric **132B** to indicate the sub-region memory access time. In a particular aspect, the sub-region access metric **192BA**, the region access metric **132B**, or both, are stored in the L2P cache **122**. In a particular aspect, the HCI **112** provides the data **206** to a component of the host device **102**, another device, or both.

The operations **200** thus enable the HCI **112** to, in response to a cache hit resulting from the cache lookup **202** in the L2P cache **122**, retrieve the physical memory address **142BAA** from the L2P cache **122** instead of requesting a mapping of the logical memory address **140BAA** from the device controller **162**. The HCI **112** also updates the sub-region access metric **192BA** and the region access metric **132B**. A technical advantage of retrieving the physical memory address **142BAA** from the L2P cache **122** can include reducing a read latency associated with reading the data **206** from the physical memory address **142BAA**, as compared to requesting a mapping of the logical memory address **140BAA** from the device controller **162** prior to sending the read request **204** to the device controller **162**. A technical advantage of maintaining the access metrics can include using the access metrics to eviction and retention decision to reduce a likelihood of cache misses.

FIG. 3 is a ladder diagram of an illustrative aspect of operations **300** associated with host device caching of address mappings of a flash memory device, in accordance

14

with some examples of the present disclosure. In a particular aspect, one or more of the operations **300** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the flash memory **164**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

In the example illustrated in FIG. 3, a memory access is to be performed of the logical memory address **140BAA** included in the sub-region **182BA** of the region **152B**, the region **152B** is not stored in the L2P cache **122**, and the sub-region **182BA** is stored in the cache **124**. The operations **300** include a cache hit **370** prior to the data access **272**.

The cache hit **370** includes the HCI **112**, in response to determining that the logical memory address **140BAA** does not match logical addresses of any of the regions **152** stored in the L2P cache **122**, detecting a cache miss **304** resulting from the cache lookup **202** of the logical memory address **140BAA** in the L2P cache **122**.

The HCI **112**, in response to detecting the cache miss **304**, performs a cache lookup **306** of the logical memory address **140BAA** in the cache **124**. The HCI **112** detects a cache hit in response to determining that the logical memory address **140BAA** matches (e.g., is included in) the logical memory addresses **140BA** of the sub-region **182BA** stored in the cache **124**. The HCI **112** reads the physical memory address **142BAA** associated with the logical memory address **140BAA** from the cache **124**. The HCI **112** performs the data access **272** of the physical memory address **142BAA**, as described with reference to FIG. 2.

The operations **300** thus enable the HCI **112** to, in response to a cache hit resulting from the cache lookup **306** in the cache **124**, retrieve the physical memory address **142BAA** from the cache **124** instead of requesting a mapping of the logical memory address **140BAA** from the device controller **162**. A technical advantage of retrieving the physical memory address **142BAA** from the cache **124** can include reducing a read latency associated with reading the data **206** from the physical memory address **142BAA**, as compared to requesting a mapping of the logical memory address **140BAA** from the device controller **162** prior to sending the read request **204** to the device controller **162**.

FIG. 4 is a ladder diagram of an illustrative aspect of operations **400** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **400** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the L2P cache **172**, the flash memory **164**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

In the example illustrated in FIG. 4, a memory access is to be performed of the logical memory address **140BAA** included in the sub-region **182BA** of the region **152B**, the region **152B** is not stored in the L2P cache **122**, the sub-region **182BA** is not stored in the cache **124**, and the L2P cache **122** is not full. The operations **400** include a cache miss **470**, a cache analysis **472**, and a sub-region activation **474** prior to the L2P cache hit **270** and the data access **272**.

The cache miss **470** includes the HCI **112**, in response to determining that the logical memory address **140BAA** does not match logical addresses of any of the sub-regions **182** stored in the cache **124**, detecting a cache miss **404** resulting from the cache lookup **306** of the logical memory address **140BAA** in the cache **124**.

The cache analysis **472** includes the HCI **112**, at block **450**, determining that the L2P cache **122** is not full. For example, the HCI **112** determines that the L2P cache **122** has

15

space available to store at least one region 152. The sub-region activation 474 includes the HCI 112 generating an activate request 412 to activate the sub-region 182BA that includes the logical memory address 140BAA and sending the activate request 412 to the device controller 162. For example, the HCI 112 generates the activate request 412 in response to the cache miss 470 and determining that the L2P cache 122 is not full.

The device controller 162, in response to receiving the activate request 412, sends a read request 414 to the L2P cache 172 to read the region 152B that includes the sub-region 182BA. The device controller 162 retrieves the region 152B from the L2P cache 172 and provides the region 152B to the HCI 112. In a particular implementation, the device controller 162, at block 452, activates the region 152B in response to receiving the activate request 412 from the HCI 112 and receiving the region 152B from the L2P cache 172. For example, the device controller 162 maintains a record of activated regions at the host device 102 and updates the record to indicate that the region 152B is activated.

The HCI 112 provides the region 152B in conjunction with an add request 416 to the L2P cache 122. For example, the HCI 112 stores the region 152B in the L2P cache 122. In a particular aspect, the HCI 112, subsequent to storing the region 152B in the L2P cache 122, updates the region access metric 132B of the region 152B. For example, the region access metric 132 indicates a time at which the region 152B is added to the L2P cache 122.

The operations 400 thus enable the HCI 112 to, in response to the cache miss 470, retrieve the region 152B from the L2P cache 172 to store in the L2P cache 122 instead of requesting a mapping of the logical memory address 140BAA from the device controller 162. A technical advantage of retrieving the entire region instead of the mapping of a single logical memory address can include reducing a read latency associated with subsequent memory accesses of logical addresses included in the region.

FIG. 5 is a ladder diagram of an illustrative aspect of operations 500 associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations 500 are performed by the HCI 112, the L2P cache 122, the cache 124, the host device 102, the device controller 162, the L2P cache 172, the flash memory 164, the flash memory device 104, the system 100 of FIG. 1, or a combination thereof.

In the example illustrated in FIG. 5, a memory access is to be performed of the logical memory address 140BAA included in the sub-region 182BA of the region 152B, the region 152B is not stored in the L2P cache 122, the sub-region 182BA is not stored in the cache 124, the L2P cache 122 is full, and the cache 124 is not full. The operations 500 include a cache analysis 572 subsequent to the cache miss 470 and prior to the sub-region activation 474, the L2P cache hit 270, and the data access 272.

The cache analysis 572 includes the HCI 112, at block 550, determining that the L2P cache 122 is full. For example, the HCI 112 determines that available space of the L2P cache 122 is less than a size of a region 152. The HCI 112, in response to the cache miss 470 and determining that the L2P cache 122 is full, makes space in the L2P cache 122 prior to sending the activate request 412 to the device controller 162. For example, the HCI 112, at block 552, identifies a region to deactivate (e.g., remove from the L2P cache 122) and identifies a sub-region of the identified region to retain in the cache 124. To illustrate, the HCI 112

16

selects the region 152A to deactivate based on determining that the region access metric 132A of the region 152A satisfies the region eviction criterion 184, and selects the sub-region 182AA of the region 152A based on determining that a sub-region access metric 192AA of the sub-region 182AA satisfies the sub-region retention criterion 186.

In a particular implementation, the region access metric 132A of the region 152A is based on a timestamp 562A, sub-region access metrics 192A of the sub-regions 182A, or a combination thereof. In a particular aspect, a sub-region access metric 192 of a sub-region 182 includes a counter 592 that indicates a total count of memory accesses to logical memory addresses of the sub-region 182. For example, the sub-region access metric 192AA includes a counter 592AA that indicates a total count of memory accesses to logical memory addresses of the sub-region 182AA. As another example, the sub-region access metric 192AB includes a counter 592AB that indicates a total count of memory accesses to logical memory addresses of the sub-region 182AB.

In a particular aspect, the region access metric 132A indicates a total count of memory accesses to logical memory addresses of the region 152A. For example, the total count of memory accesses to logical memory addresses of the region 152A correspond to a sum of the total counts of memory accesses to logical memory addresses of the sub-regions 182A.

In a particular aspect, the timestamp 562A indicates a time at which the region 152A is added to the L2P cache 122, a most recent memory access to any logical memory address of the region 152A, or both. In an example, a sub-region access metric 192 of a sub-region 182 includes a timestamp that indicates a sub-region memory access time of a most recent memory access to any logical memory address of the sub-region 182. For example, the sub-region access metric 192AA includes a timestamp (not shown in FIG. 5) that indicates a first sub-region memory access time of a most recent memory access to any logical memory address of the sub-region 182AA. As another example, the sub-region access metric 192AB includes a timestamp that indicates a second sub-region memory access time of a most recent memory access to any logical memory address of the sub-region 182AB. In a particular aspect, the timestamp 562A of the region access metric 132A indicates a most recent time of the sub-region memory access times indicated by the sub-region access metrics 192A.

In a particular aspect, the region access metric 132A of the region 152A is stored in the L2P cache 122. Similarly, in some aspects, one or more additional regions 152 and corresponding region access metrics 132 are stored in the L2P cache 122. For example, the region 152C and a region access metric 132C (e.g., counters 592C, a timestamp 562C, or both) of the region 152C are stored in the L2P cache 122.

According to some implementations, the HCI 112 determines that the region access metric 132A of the region 152A satisfies the region eviction criterion 184 based at least in part on a comparison of the region access metric 132A and one or more thresholds. For example, the HCI 112 determines that the region access metric 132A of the region 152A satisfies the region eviction criterion 184 based at least in part on determining that the total count of memory accesses (indicated by the counters 592A) of the region 152A is less than an access count threshold, that the timestamp 562A indicates an add time (that the region 152A is added to the L2P cache 122) that is earlier than an add time threshold, that the timestamp 562A indicates a most recent memory

17

access time of the region **152A** that is earlier than an access time threshold, or a combination thereof.

According to some implementations, the HCI **112** determines that the region access metric **132A** of the region **152A** satisfies the region eviction criterion **184** based at least in part on a comparison of the region access metric **132A** of the region **152A** to the region access metrics **132** of other regions **152** stored in the L2P cache **122**. For example, the HCI **112** determines that the region access metric **132A** of the region **152A** satisfies the region eviction criterion **184** based at least in part on determining that the counters **592A** indicate (e.g., a sum of the counter values has a lowest value indicating) the least count of accesses among counts of accesses indicated by the counters **592** of each of the regions **152** stored in the L2P cache **122**. As another example, the HCI **112** determines that the region access metric **132A** of the region **152A** satisfies the region eviction criterion **184** based at least in part on determining that the timestamp **562A** indicates an add time that is earliest among add times indicated by timestamps **562** of the regions **152** stored in the L2P cache **122**. As yet another example, the HCI **112** determines that the region access metric **132A** of the region **152A** satisfies the region eviction criterion **184** based at least in part on determining that the timestamp **562A** indicates a most recent access time that is earliest among most recent access times indicated by the timestamps **562** of the regions **152** stored in the L2P cache **122**.

The HCI **112**, in response to determining that the region **152A** is selected to be deactivated, selects the sub-region **182AA** of the region **152A** based on determining that a sub-region access metric **192AA** of the sub-region **182AA** satisfies the sub-region retention criterion **186**. According to some implementations, the HCI **112**, based at least in part on determining that the sub-region access metric **192AA** (e.g., a count of accesses indicated by the counter **592AA**) exceeds an access count threshold, determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**. According to some implementations, the HCI **112**, based at least in part on determining that the sub-region access metric **192AA** (e.g., a count of accesses indicated by the counter **592AA**) has a highest access count value (e.g., highest counter value) among the access count values indicated by the sub-region access metrics **192A** of other sub-regions **182A** of the region **152A**, determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**.

According to some implementations, the HCI **112**, based at least in part on determining that the sub-region access metric **192AA** exceeds an access time threshold, determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**. For example, the HCI **112**, based at least in part on determining that the sub-region access metric **192AA** (e.g., a timestamp) indicates a most recent memory access time to any logical memory address of the sub-region **182AA** that is later than an access time threshold (e.g., indicating that an address in the sub-region **182AA** was used more recently than a time indicated by the access time threshold), determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**.

According to some implementations, the HCI **112**, based at least in part on a comparison of an access time indicated by the sub-region access metric **192AA** and access times indicated by sub-region access metrics **192A** of other sub-regions **182A** of the region **152A**, determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**. For example, the HCI **112**, based at

18

least in part on determining that the sub-region access metric **192AA** (e.g., a timestamp) indicates a most recent memory access time to any logical memory address of the sub-region **182AA** that is later than most recent memory access times indicated by the sub-region access metrics **192A** of other sub-regions **182A**, determines that the sub-region access metric **192AA** satisfies the sub-region retention criterion **186**.

The HCI **112** determines, at block **554**, that the cache **124** is not full. For example, the HCI **112**, in response to determining that the cache **124** has space to store at least one sub-region, determines that the cache **124** is not full. The HCI **112**, in response to determining that the cache **124** is not full, performs region deactivation **574** to deactivate the region **152A**. Performing the region deactivation **574** includes sending a copy request **506** to the L2P cache **122** to copy the sub-region **182AA** from the L2P cache **122** to the cache **124**. The HCI **112** sends a remove request **508** to the L2P cache **122** to remove the region **152A** from the L2P cache **122**.

According to some implementations, the HCI **112** sends a deactivate request **510** to the device controller **162** to deactivate the region **152A**. The device controller **162**, at block **556**, deactivates the region **152A** in response to receiving the deactivate request **510**. For example, the device controller **162** updates activation data to indicate that the region **152A** is deactivated. The HCI **112** performs the sub-region activation **474** after removing the region **152A** from the L2P cache **122**.

The operations **500** thus enable the HCI **112** to, in response to the cache miss **470** and determining that the L2P cache **122** is full, evict the region **152A** from the L2P cache **122** based on the region access metric **132A** of the region **152A** to reduce a likelihood of cache misses in the L2P cache **122**. The HCI **112** retains the sub-region **182AA** of the region **152A** in the cache **124**. A technical advantage of retaining the sub-region **182AA** in the cache **124** can include reducing a read latency associated with reading from logical memory addresses of the sub-region **182AA** after evicting the region **152A** from the L2P cache **122**.

FIG. 6 is a ladder diagram of an illustrative aspect of operations **600** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **600** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the L2P cache **172**, the flash memory **164**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

In the example illustrated in FIG. 6, a memory access is to be performed of the logical memory address **140BAA** included in the sub-region **182BA** of the region **152B**, the region **152B** is not stored in the L2P cache **122**, the sub-region **182BA** is not stored in the cache **124**, the L2P cache **122** is full, and the cache **124** is full. The operations **600** include a cache analysis **672** subsequent to the cache miss **470** and prior to the sub-region activation **474**, the L2P cache hit **270**, and the data access **272**.

The cache analysis **672** includes the HCI **112**, at block **650**, determining that the cache **124** is full. For example, the HCI **112**, in response to determining that an available space of the cache **124** is less than a size of a sub-region **182**, determines that the cache **124** is full. The HCI **112**, in response to the cache miss **470**, determining that the L2P cache **122** is full, and that the cache **124** is full, makes space in the cache **124** to make space in the L2P cache **122** prior to sending the activate request **412** to the device controller

162. For examples, the HCI 112, at block 652, identifies a sub-region 182 to remove from the cache 124. To illustrate, the HCI 112 selects a sub-region 182EB stored in the cache 124 in response to determining that a sub-region access metric 192EB of the sub-region 182EB satisfies the sub-region eviction criterion 188.

In an example, the sub-region access metric 192EB is based on a counter 592EB, a timestamp 662EB, or both. In a particular implementation, the counter 592EB indicates a total count of memory accesses to logical memory addresses of the sub-region 182EB during a particular time period. In a particular implementation, the timestamp 662EB indicates an add time that the sub-region 182EB is added to (e.g., stored in) the cache 124, an access time of a most recent access to any logical memory address of the sub-region 182EB, or both.

According to some implementations, the HCI 112 determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188 based on a comparison of the sub-region access metric 192EB to one or more thresholds. For example, the HCI 112, based at least in part on determining that the sub-region access metric 192EB (e.g., a count of memory accesses indicated by the counter 592EB) is less than a threshold access count, determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188. In another example, the HCI 112, based at least in part on determining that a time indicated by the sub-region access metric 192EB (e.g., the add time, the access time, or both indicated by the timestamp 662EB) is earlier than an access time threshold, determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188.

According to some implementations, the HCI 112 determines that the sub-region access metric 192EB of the sub-region 182EB satisfies the sub-region eviction criterion 188 based on a comparison of the sub-region access metric 192EB to sub-region access metrics 192 of other sub-regions 182 stored in the cache 124. For example, the HCI 112, based at least in part on determining that the sub-region access metric 192EB (e.g., a count of accesses indicated by the counter 592EB) has a lowest access count value (e.g., lowest counter value) among the access count values indicated by the sub-region access metrics 192 of other sub-regions 182 stored in the cache 124, determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188. As another example, the HCI 112, based at least in part on determining that the sub-region access metric 192EB (e.g., the add time, the access time, or both) indicates an earliest time among the times indicated by the sub-region access metrics 192 of other sub-regions 182 stored in the cache 124, determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188. To illustrate, the HCI 112, in response to determining that the sub-region access metric 192EB indicates that the sub-region 182EB is a first sub-region (that has the earliest add time of the sub-regions 182 included in the cache 124) to be added to the cache 124, determines that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188.

The HCI 112, in response to determining that the sub-region access metric 192EB satisfies the sub-region eviction criterion 188, sends a remove request 608 to the cache 124 to remove the sub-region 182EB from the cache 124. The HCI 112 performs the region deactivation 574 after removing the sub-region 182EB from the cache 124.

It should be understood that evicting a single sub-region from the cache 124 is provided as an illustrative example. In

other examples, multiple sub-regions 182A of the region 152A may be retained in the cache 124 and one or more sub-regions 182 may be evicted from the cache 124 to make space for one or more of the sub-regions 182A to be retained. For example, the sub-regions 182 stored in the cache 124 may be ranked based on the sub-region access metrics 192 and one or more of the sub-regions 182 may be evicted from the cache 124 based on the rankings.

The operations 600 thus enable the HCI 112 to, in response to the cache miss 470, determining that the L2P cache 122 is full, and determining that the cache 124 is full, evict the sub-region 182EB from the cache 124 based on an access metric to reduce a likelihood of cache misses. The HCI 112 replaces the sub-region 182EB with a sub-region 182AA of the region 152A in the cache 124. A technical advantage of retaining the sub-region 182AA in the cache 124 can include reducing a read latency associated with reading from logical memory addresses of the sub-region 182AA after evicting the region 152A from the L2P cache 122.

FIG. 7 is a ladder diagram of an illustrative aspect of operations 700 associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations 700 are performed by the HCI 112, the L2P cache 122, the cache 124, the host device 102, the device controller 162, the flash memory device 104, the system 100 of FIG. 1, or a combination thereof.

The device controller 162 sends a notification 702 to the HCI 112 to inactivate the region 152A. For example, the device controller 162 sends the notification 702 in response to determining that one or more sub-regions 182A of the region 152A have been updated at the flash memory 164. To illustrate, in some examples, multiple sub-regions 182A of the region 152A could be updated due to reshuffling of the data to different physical locations in the flash memory 164. In some examples, sub-region(s) 182 can be updated because one or more logical memory address are deallocated, e.g., due to data deletions.

The HCI 112, in response to receiving the notification 702, performs a region deactivation 774. Performing the region deactivation 774 includes sending the remove request 508 to the L2P cache 122 to remove the region 152A from the L2P cache 122. The L2P cache 122, in response to receiving the remove request 508 and determining that the region 152A is stored at the L2P cache 122, removes the region 152A from the L2P cache 122. The HCI 112 sends a remove request 708 to the cache 124 to remove any sub-regions 182A of the region 152A that are stored in the cache 124. The cache 124, in response to receiving the remove request 708, removes any sub-regions 182A of the region 152A that are stored at the cache 124.

The notification 702 could correspond to the region 152A including one or more invalid (e.g., outdated) mappings. The HCI 112 does not retain any of the sub-regions 182A in the cache 124 during the region deactivation 774 of the region 152A to prevent retaining invalid mappings. The operations 700 thus enable the device controller 162 to remove mappings from the cache 124 in addition to the L2P cache 122.

FIG. 8 is a ladder diagram of an illustrative aspect of operations 800 associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations 800 are performed by the HCI 112, the L2P cache 122, the cache 124, the host

21

device **102**, the device controller **162**, the L2P cache **172**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

The device controller **162** sends a notification **802** to the HCI **112** to inactivate the region **152A** and activate the sub-region **182BA**. For example, the device controller **162** sends the notification **802** in response to determining that one or more sub-regions **182** of the region **152A** have been updated (e.g., moved or deleted) at the flash memory **164**, and that the sub-region **182BA** is recently added to the flash memory **164**.

The HCI **112**, in response to receiving the notification **802**, performs the region deactivation **774** of the region **152A**, as described with reference to FIG. 7. The HCI **112** performs cache analysis **872** after performing the region deactivation **774**. The cache analysis **872** corresponds to the cache analysis **472** of FIG. 4 when the L2P cache **122** is not full. The cache analysis **872** corresponds to the cache analysis **572** of FIG. 5 when the L2P cache **122** is full and the cache **124** is not full. The cache analysis **872** corresponds to the cache analysis **672** of FIG. 6 when the L2P cache **122** is full and the cache **124** is full.

Subsequent to the cache analysis **872**, the L2P cache **122** includes space for the region **152B** that includes the sub-region **182BA**. In some examples, a region **152** is evicted from the L2P cache **122** to make space for the region **152B** and one or more sub-regions **182** of the evicted region **152** are retained in the cache **124**.

The HCI **112**, after performing the cache analysis **872**, performs the sub-region activation **474** of the sub-region **182BA**, as described with reference to FIG. 4. In some implementations, the device controller **162**, prior to sending the notification **802** to the HCI **112**, updates activation data to indicate that the region **152A** is inactivated, the region **152B** that includes the sub-region **182BA** is activated, or both. The operations **800** thus enable the device controller **162** to remove mappings of the region **152A** from the cache **124** in addition to the L2P cache **122**, and to store mappings of the region **152B** in the L2P cache **122**.

FIG. 9 is a ladder diagram of an illustrative aspect of operations **900** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **900** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the L2P cache **172**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

The device controller **162** sends a notification **902** to the HCI **112** to activate the sub-region **182BA**. For example, the device controller **162** sends the notification **902** in response to determining that the sub-region **182BA** is recently added to the flash memory **164**.

The HCI **112**, in response to receiving the notification **902**, performs the cache analysis **872**, as described with reference to FIG. 8. Subsequent to the cache analysis **872**, the L2P cache **122** includes space for the region **152B** that includes the sub-region **182BA**. In some examples, a region **152** is evicted from the L2P cache **122** to make space for the region **152B** and one or more sub-regions **182** of the evicted region **152** are retained in the cache **124**.

The HCI **112**, after performing the cache analysis **872**, performs the sub-region activation **474** of the sub-region **182BA**, as described with reference to FIG. 4. In some implementations, the device controller **162**, prior to sending the notification **902** to the HCI **112**, updates activation data to indicate that the region **152B** that includes the sub-region

22

182BA is activated. The operations **900** thus enable the device controller **162** to store mappings of the region **152B** in the L2P cache **122**.

FIG. 10 is a ladder diagram of an illustrative aspect of operations **1000** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **1000** are performed by the HCI **112**, the L2P cache **122**, the host device **102**, the device controller **162**, the L2P cache **172**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

The device controller **162** sends a notification **1002** to the HCI **112** to update the sub-region **182BA**. For example, the device controller **162** sends the notification **1002** in response to determining that one or more mappings of the sub-region **182BA** are updated at the L2P cache **172** of the flash memory device **104**.

The HCI **112**, in response to receiving the notification **1002**, sends a read request **1012** to read the sub-region **182BA** to the device controller **162**. The device controller **162**, in response to receiving the read request **1012**, sends a read request **1014** to read the sub-region **182BA** to the L2P cache **172**. For example the device controller **162** reads the sub-region **182BA** from the L2P cache **172**. The device controller **162** provides the sub-region **182BA** to the HCI **112**.

The HCI **112** sends an update request **1016** to update the sub-region **182BA** to the L2P cache **122**. The L2P cache **122**, in response to receiving the update request **1016** and determining that the sub-region **182BA** is stored in the L2P cache **122**, replaces the sub-region **182BA** store in the L2P cache **122** with the sub-region **182BA** received from the HCI **112** (or the device controller **162**). The operations **1000** thus enable the device controller **162** to update mappings of the sub-region **182BA** stored in the L2P cache **122**.

FIG. 11 is a ladder diagram of an illustrative aspect of operations **1100** associated with host device caching of address mappings of a flash memory device, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations **1100** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the L2P cache **172**, the flash memory device **104**, the system **100** of FIG. 1, or a combination thereof.

The L2P cache **122**, in response to receiving the update request **1016** and determining that the sub-region **182BA** is not stored in the L2P cache **122**, provides an indication of a cache miss **1104** to the HCI **112**. The HCI **112**, in response to detecting the cache miss **1104**, sends an update request **1118** to update the sub-region **182BA** to the cache **124**. The cache **124**, in response receiving the update request **1118** and determining that the sub-region **182BA** is stored at the cache **124**, replaces the sub-region **182BA** store at the cache **124** with the sub-region **182BA** received from the HCI **112** (or the device controller **162**). The operations **1100** thus enable the device controller **162** to update mappings of the sub-region **182BA** stored in the cache **124**.

FIG. 12 is a diagram of a particular implementation of a method **1200** of host device caching of address mappings of a flash memory device that may be performed by the device of FIG. 1, in accordance with some examples of the present disclosure. In a particular aspect, one or more of the operations of the method **1200** are performed by the HCI **112**, the L2P cache **122**, the cache **124**, the host device **102**, the device controller **162**, the L2P cache **172**, the device con-

23

troller 162, the flash memory 164, the flash memory device 104, the system 100 of FIG. 1, or a combination thereof.

The method 1200 includes, at block 1202, initiation of a read request from a file system. For example, the HCI 112 of FIG. 1 receives a memory access request (e.g., a read request) from a file system of the host device 102. In a particular aspect, the memory access request indicates a logical memory address 140BAA of data to be accessed.

The method 1200 also includes, at block 1204, retrieving host controller commands. For example, the HCI 112 retrieves host controller commands that are supported (e.g., activated) at the HCI 112.

The method 1200 further includes, at block 1206, determining whether HPB is supported. For example, the HCI 112 determines whether retrieved host controller commands indicate that HPB is supported at the host device 102 for the flash memory device 104.

The method 1200 includes, in response to determining, at block 1206, that HPB is not supported, performing legacy mode operations 1208. Performing the legacy mode operations 1208 includes accessing, at block 1210, an address of a memory controller. For example, the HCI 112, in response to determining that HPB is not supported, retrieves an address of the device controller 162.

Performing the legacy mode operations 1208 also includes refreshing, at block 1212, entries using SRAM. For example, the HCI 112 sends a mapping request to the device controller 162 to retrieve the physical memory address 142BAA associated with the logical memory address 140BAA from the L2P address mapping table 174 stored in the L2P cache 172 (e.g., SRAM).

Performing the legacy mode operations 1208 further includes reading, at block 1214, the flash memory data. For example, the HCI 112 sends a read request to the device controller 162 for the physical memory address 142BAA and receives the data 206 from the device controller 162 retrieved from the physical memory address 142BAA of the flash memory 164.

The method 1200 includes, in response to determining, at block 1206, that HPB is supported, determining, at block 1216, whether a L2P cache miss is detected. For example, the HCI 112, in response to determining that HPB is supported, performs a cache lookup for the logical memory address 140BAA in the L2P cache 122.

The method 1200 includes, in response to determining, at block 1216, that a L2P cache hit is detected, reading, at block 1218, the flash memory data. For example, the HCI 112, in response to detecting a cache hit resulting from the cache lookup in the L2P cache 122, performs the data access 272, as described with reference to FIG. 2. To illustrate, the HCI 112 sends a read request 204 for the physical memory address 142BAA to the device controller 162 and receives the data 206 from the device controller 162. The data 206 is retrieved from the physical memory address 142BAA of the flash memory 164.

The method 1200 includes, in response to determining, at block 1216, that a L2P cache miss is detected, determining, at block 1220, whether the L2P cache is full. For example, the HCI 112, in response to detecting the cache miss 304, determines whether the L2P cache 122 is full, as described with reference to FIGS. 4 and 5.

The method 1200 includes, in response to determining, at block 1220, that the L2P cache is not full, adding, at block 1222, a region to the L2P cache and proceeding to block 1218. For example, the HCI 112, in response to determining that the L2P cache 122 is not full, performs the sub-region activation 474 and the data access 272, as described with

24

reference to FIG. 4. To illustrate, the HCI 112 adds the region 152B to the L2P cache 122, sends a read request 204 for the physical memory address 142BAA to the device controller 162, and receives the data 206 from the device controller 162.

The method 1200 includes, in response to determining, at block 1220, that the L2P cache is full, selecting, at block 1224, a region to be removed from the L2P cache. For example, the HCI 112, in response to determining that the L2P cache 122 is full, selects the region 152A to be removed from the L2P cache 122, as described with reference to FIG. 5.

The method 1200 also includes copying, at block 1226, a selected sub-region of the selected region to an RIR cache. For example, the HCI 112 copies the sub-region 182AA to the cache 124 (e.g., an RIR cache).

The method 1200 also includes removing, at block 1228, the selected region from the L2P cache and proceeding to the block 1222. For example, the HCI 112 removes the region 152A from the L2P cache 122 and adds the region 152B to the L2P cache 122.

The method 1200 enables the sub-region 182AA to be available in the cache 124 subsequent to eviction of the region 152A from the L2P cache 122. A technical advantage of having the sub-region 182AA in the cache 124 can include reduced memory access latency associated with subsequent memory accesses of logical memory addresses of the sub-region 182AA.

The method 1200 of FIG. 12 may be implemented by a field-programmable gate array (FPGA) device, an application-specific integrated circuit (ASIC), a processing unit such as a central processing unit (CPU), a digital signal processor (DSP), a controller, another hardware device, firmware device, or any combination thereof. As an example, the method 1200 of FIG. 12 may be performed by a processor that executes instructions, such as described with reference to FIG. 14.

Referring to FIG. 13, a particular implementation of a method 1300 of host device caching of address mappings of a flash memory device is shown. In a particular aspect, one or more operations of the method 1300 are performed by at least one of the HCI 112, the L2P cache 122, the cache 124, the host device 102, the device controller 162, the L2P cache 172, the flash memory 164, the flash memory device 104, the system 100 of FIG. 1, or a combination thereof.

The method 1300 includes, at block 1302, determining, at a host device coupled to a flash memory device, that a particular region of a logical-to-physical (L2P) address mapping table is to be removed from a L2P cache of the host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of the flash memory device. For example, the host device 102 is coupled to the flash memory device 104. The HCI 112 of the host device 102 determines that the region 152A of the L2P address mapping table 174 is to be removed from the L2P cache 122 of the host device 102, as described with reference to FIGS. 1 and 5. The L2P address mapping table 174 is configured to include mappings between the logical memory addresses 140 and the physical memory addresses 142 of the flash memory 164 of the flash memory device 104.

The method 1300 also includes, at block 1304, identifying, at the host device, a particular sub-region of the particular region having an access metric that satisfies a retention criterion. For example, the HCI 112 of the host device 102 identifies the sub-region 182AA of the region 152A

having the sub-region access metric **192AA** that satisfies the sub-region retention criterion **186**, as described with reference to FIGS. **1** and **5**.

The method **1300** further includes, at block **1306**, storing the particular sub-region into a second cache of the host device. For example, the HCI **112** stores the sub-region **182AA** in the cache **124** of the host device **102**, as described with reference to FIGS. **1** and **5**.

The method **1300** also includes, at block **1308**, removing the particular region from the L2P cache. For example, the HCI **112** removes the region **152A** from the L2P cache **122**, as described with reference to FIGS. **1** and **5**.

The method **1300** enables the sub-region **182AA** to remain available at the host device **102** in the cache **124** subsequent to eviction of the region **152A** from the L2P cache **122**. A technical advantage of the method **1300** can include reducing memory access latency associated with accessing memory address mappings of logical memory addresses included in the sub-region **182AA** subsequent to eviction of the region **152A** from the L2P cache **122**.

The method **1300** of FIG. **13** may be implemented by a FPGA device, an ASIC, a processing unit such as a CPU, a DSP, a controller, another hardware device, firmware device, or any combination thereof. As an example, the method **1300** of FIG. **13** may be performed by a processor that executes instructions, such as described with reference to FIG. **14**.

Referring to FIG. **14**, a block diagram of a particular illustrative implementation of a device is depicted and generally designated **1400**. In various implementations, the device **1400** may have more or fewer components than illustrated in FIG. **14**. In an illustrative implementation, the device **1400** may correspond to the host device **102**. In an illustrative implementation, the device **1400** may perform one or more operations described with reference to FIGS. **1-13**.

In a particular implementation, the device **1400** includes a processor **1406** (e.g., a CPU). The device **1400** may include one or more additional processors **1410** (e.g., one or more DSPs). The processors **1410** may include a speech and music coder-decoder (CODEC) **1408** that includes a voice coder (“vocoder”) encoder **1436**, a vocoder decoder **1438**, or both. The processors **1410** may include the HCI **112** that is configured to be coupled to the flash memory device **104**.

The device **1400** may include the system memory **114** and a CODEC **1434**. The system memory **114** may include instructions **1456**, that are executable by the one or more additional processors **1410** (or the processor **1406**) to implement the functionality described with reference to the HCI **112**. The system memory **114** may include the L2P cache **122**, the cache **124**, or both. The device **1400** may include a modem **1470** coupled, via a transceiver **1450**, to an antenna **1452**.

The device **1400** may include a display **1428** coupled to a display controller **1426**. One or more speakers **1492** and one or more microphones **1494** may be coupled to the CODEC **1434**. The CODEC **1434** may include a digital-to-analog converter (DAC) **1402**, an analog-to-digital converter (ADC) **1404**, or both. In a particular implementation, the CODEC **1434** may receive analog signals from the microphone(s) **1494**, convert the analog signals to digital signals using the analog-to-digital converter **1404**, and provide the digital signals to the speech and music codec **1408**. The speech and music codec **1408** may process the digital signals. In a particular implementation, the speech and music codec **1408** may provide digital signals to the CODEC **1434**. The CODEC **1434** may convert the digital

signals to analog signals using the digital-to-analog converter **1402** and may provide the analog signals to the speaker(s) **1492**.

In a particular implementation, the device **1400** may be included in a system-in-package or system-on-chip device **1422**. In a particular implementation, the system memory **114**, the processor **1406**, the processors **1410**, the display controller **1426**, the CODEC **1434**, and the modem **1470** are included in the system-in-package or system-on-chip device **1422**. In a particular implementation, an input device **1430**, a power supply **1444**, and the flash memory device **104** are coupled to the system-in-package or the system-on-chip device **1422**. Moreover, in a particular implementation, as illustrated in FIG. **14**, the display **1428**, the input device **1430**, the speaker(s) **1492**, the microphone(s) **1494**, the antenna **1452**, the flash memory device **104**, and the power supply **1444** are external to the system-in-package or the system-on-chip device **1422**. In a particular implementation, each of the display **1428**, the input device **1430**, the speaker(s) **1492**, the microphone(s) **1494**, the antenna **1452**, the flash memory device **104**, and the power supply **1444** may be coupled to a component of the system-in-package or the system-on-chip device **1422**, such as an interface or a controller. For example, the flash memory device **104** may be coupled to the HCI **112**.

The device **1400** may include a smart speaker, a speaker bar, a mobile communication device, a smart phone, a cellular phone, a laptop computer, a computer, a tablet, a personal digital assistant, a display device, a television, a gaming console, a music player, a radio, a digital video player, a digital video disc (DVD) player, a tuner, a camera, a navigation device, a vehicle, a headset, an augmented reality headset, a mixed reality headset, a virtual reality headset, an aerial vehicle, a home automation system, a voice-activated device, a wireless speaker and voice activated device, a portable electronic device, a car, a computing device, a communication device, an internet-of-things (IoT) device, a virtual reality (VR) device, a base station, a mobile device, or any combination thereof.

In conjunction with the described implementations, an apparatus includes means for determining that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of a host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device. For example, the means for determining that a particular region is to be removed can correspond to the HCI **112**, the host device **102**, the system **100** of FIG. **1**, the processor **1406**, the processor(s) **1410**, one or more other circuits or components configured to determine that a particular region is to be removed, or any combination thereof.

The apparatus also includes means for identifying a particular sub-region of the particular region having an access metric that satisfies a retention criterion. For example, the means for identifying a particular sub-region can correspond to the HCI **112**, the host device **102**, the system **100** of FIG. **1**, the processor **1406**, the processor(s) **1410**, one or more other circuits or components configured to identify a particular sub-region, or any combination thereof.

The apparatus further includes means for storing the particular sub-region into a second cache of the host device. For example, the means for storing can correspond to the HCI **112**, the host device **102**, the system **100** of FIG. **1**, the processor **1406**, the processor(s) **1410**, one or more other

circuits or components configured to identify a particular sub-region, or any combination thereof.

The apparatus also includes means for removing the particular region from the L2P cache. For example, the means for removing the particular region from the L2P cache can correspond to the HCI **112**, the host device **102**, the system **100** of FIG. **1**, the processor **1406**, the processor(s) **1410**, one or more other circuits or components configured to remove the particular region, or any combination thereof.

In some implementations, a non-transitory computer-readable medium (e.g., a computer-readable storage device, such as the system memory **114**) includes instructions (e.g., the instructions **1456**) that, when executed by one or more processors (e.g., the one or more processors **1410** or the processor **1406**), cause the one or more processors to determine that a particular region (e.g., the region **152A**) of a logical-to-physical (L2P) mapping table (e.g., the L2P address mapping table **174**) is to be removed from a L2P cache (e.g., the L2P cache **122**) of a host device (e.g., the host device **102**), the L2P address mapping table configured to include mappings between logical memory addresses (e.g., the logical memory addresses **140**) and physical memory addresses (e.g., the physical memory addresses **142**) of a flash memory device (e.g., the flash memory **164** of the flash memory device **104**). The instructions further cause the one or more processors to identify a particular sub-region (e.g., the sub-region **182AA**) of the particular region having an access metric (e.g., the sub-region access metric **192AA**) that satisfies a retention criterion (e.g., the sub-region retention criterion **186**). The instructions also cause the one or more processors to store the particular sub-region into a second cache (e.g., the cache **124**) of the host device. The instructions further cause the one or more processors to remove the particular region from the L2P cache.

Particular aspects of the disclosure are described below in sets of interrelated Examples:

According to Example 1, a host device includes system memory including a logical-to-physical (L2P) cache and a second cache; and a host controller interface (HCI) configured to be coupled to a flash memory device and configured to determine that a particular region of a L2P address mapping table is to be removed from the L2P cache, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of the flash memory device; identify a particular sub-region of the particular region having an access metric that satisfies a retention criterion; store the particular sub-region into the second cache; and remove the particular region from the L2P cache.

Example 2 includes the host device of Example 1, wherein the L2P cache corresponds to a Host Performance Booster (HPB) cache, and wherein the second cache is implemented in a recently inactivated region (RIR) partition of the system memory.

Example 3 includes the host device of Example 1 or Example 2, wherein the HCI is configured to, in response to a cache miss resulting from a first cache lookup at the L2P cache, perform a second cache lookup at the second cache.

Example 4 includes the host device of any of Examples 1 to 3, wherein the HCI is configured to perform a cache lookup at the second cache including comparing a logical address of data to be accessed to logical addresses corresponding to sub-regions stored in the second cache; and in response to detecting a match of the logical address to particular logical addresses of the particular sub-region

stored in the second cache, read a physical address associated with the logical address from the particular sub-region and send the physical address to the flash memory device in conjunction with a request to access the data.

Example 5 includes the host device of any of Examples 1 to 4, wherein the HCI is configured to remove a second region from the L2P cache in response to a notification from the flash memory device that the second region is to be removed.

Example 6 includes the host device of any of Examples 1 to 5, wherein the HCI is configured to determine that the particular region is to be removed in response to a cache miss resulting from a cache lookup of a logical address at the L2P cache and at the second cache.

Example 7 includes the host device of Example 6, wherein the HCI is configured to, in response to the cache miss: send the logical address to the flash memory device; receive a replacement region that includes a first sub-region associated with the logical address; and after removing the particular region from the L2P cache, store the replacement region in the L2P cache.

Example 8 includes the host device of any of Examples 1 to 7, wherein the access metric is based on a count of accesses to the particular sub-region.

Example 9 includes the host device of any of Examples 1 to 8, wherein the HCI is further configured to maintain a counter for each sub-region of the particular region, wherein the counter for a sub-region is updated each time that sub-region is accessed.

Example 10 includes the host device of any of Examples 1 to 9, wherein the HCI is configured to identify the particular sub-region responsive to determining that a counter associated with the particular sub-region has a highest counter value of counters that are associated with sub-regions of the particular region.

Example 11 includes the host device of any of Examples 1 to 10, wherein the HCI is configured to determine that the retention criterion is satisfied based on determining that the access metric exceeds an access count threshold.

Example 12 includes the host device of any of Examples 1 to 11, wherein the access metric is stored in the L2P cache.

Example 13 includes the host device of any of Examples 1 to 12, wherein the HCI is configured to responsive to determining that the second cache is full, determine that a second sub-region is to be removed from the second cache; and remove the second sub-region from the second cache, wherein the particular sub-region is stored into the second cache after removing the second sub-region from the second cache.

Example 14 includes the host device of Example 13, wherein the HCI is configured to determine that the second sub-region is to be removed responsive to determining that the second sub-region has a second access metric that satisfies a sub-region eviction criterion.

Example 15 includes the host device of Example 14, wherein the second access metric includes a first sub-region added to the second cache, a less than threshold access count, or both.

According to Example 16, a method includes at a host device coupled to a flash memory device: determining that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of the host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of the flash memory device; identifying a particular sub-region of the particular region having an access metric that satisfies a retention criterion; storing the

particular sub-region into a second cache of the host device; and removing the particular region from the L2P cache.

Example 17 includes the method of Example 16, wherein the L2P cache corresponds to a Host Performance Booster (HPB) cache in a system memory of the host device, and wherein the second cache is implemented in a recently inactivated region (RIR) partition of the system memory of the host device.

Example 18 includes the method of Example 16 or Example 17, further comprising, in response to a cache miss resulting from a first cache lookup at the L2P cache, performing a second cache lookup at the second cache.

Example 19 includes the method of any of Examples 16 to 18 and further includes performing a cache lookup at the second cache including: comparing a logical address of data to be accessed to logical addresses corresponding to sub-regions stored in the second cache; and in response to detecting a match of the logical address to the particular sub-region stored in the second cache, reading a physical address associated with the logical address from the particular sub-region and sending the physical address to the flash memory device in conjunction with a request to access the data.

Example 20 includes the method of any of Examples 16 to 19, and further includes removing a second region from the L2P cache responsive to a notification from the flash memory device that the second region is to be removed.

Example 21 includes the method of any of Examples 16 to 20, wherein the particular region is to be removed is determined responsive to a cache miss resulting from a cache lookup of a logical address at the L2P cache and at the second cache, and further comprising, in response to the cache miss: sending the logical address to the flash memory device; receiving a replacement region that includes a first sub-region associated with the logical address; and after removing the particular region from the L2P cache, storing the replacement region in the L2P cache.

Example 22 includes the method of any of Examples 16 to 21, wherein the access metric is based on a count of accesses to the particular sub-region.

Example 23 includes the method of any of Examples 16 to 22 and further includes maintaining a counter for each sub-region of the particular region, wherein the counter for a sub-region is updated each time that sub-region is accessed.

Example 24 includes the method of any of Examples 16 to 23, wherein the particular sub-region is identified responsive to determining that a counter associated with the particular sub-region has a highest counter value of counters that are associated with sub-regions of the particular region.

Example 25 includes the method of any of Examples 16 to 24, wherein satisfying the retention criterion includes determining that the access metric exceeds an access count threshold.

Example 26 includes the method of any of Examples 16 to 25, wherein the access metric is stored in the L2P cache.

Example 27 includes the method of any of Examples 16 to 26, further includes responsive to determining that the second cache is full, determining that a second sub-region is to be removed from the second cache; and removing the second sub-region from the second cache, wherein the particular sub-region is stored into the second cache after removing the second sub-region from the second cache.

Example 28 includes the method of Example 27, wherein the second sub-region is to be removed is determined

responsive to determining that the second sub-region has a second access metric that satisfies a sub-region eviction criterion.

Example 29 includes the method of Example 28, wherein the second access metric includes a first sub-region added to the second cache, a less than threshold access count, or both.

According to Example 30, a device includes: a memory configured to store instructions; and a processor configured to execute the instructions to perform the method of any of Example 16 to 29.

According to Example 31, a non-transitory computer-readable medium stores instructions that, when executed by a processor, cause the processor to perform the method of any of Example 16 to Example 29.

According to Example 32, an apparatus includes means for carrying out the method of any of Example 16 to Example 29.

According to Example 33, a non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to determine that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of a host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device; identify a particular sub-region of the particular region having an access metric that satisfies a retention criterion; store the particular sub-region into a second cache of the host device; and remove the particular region from the L2P cache.

According to Example 34, an apparatus includes means for determining that a particular region of a logical-to-physical (L2P) mapping table is to be removed from a L2P cache of a host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device; means for identifying a particular sub-region of the particular region having an access metric that satisfies a retention criterion; means for storing the particular sub-region into a second cache of the host device; and means for removing the particular region from the L2P cache.

Those of skill would further appreciate that the various illustrative logical blocks, configurations, modules, circuits, and algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software executed by a processor, or combinations of both. Various illustrative components, blocks, configurations, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or processor executable instructions depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, such implementation decisions are not to be interpreted as causing a departure from the scope of the present disclosure.

The steps of a method or algorithm described in connection with the implementations disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), registers, hard disk, a removable disk, a compact disc read-only memory (CD-

31

ROM), or any other form of non-transient storage medium known in the art. An exemplary storage medium is coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit (ASIC). The ASIC may reside in a computing device or a user terminal. In the alternative, the processor and the storage medium may reside as discrete components in a computing device or user terminal.

The previous description of the disclosed aspects is provided to enable a person skilled in the art to make or use the disclosed aspects. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other aspects without departing from the scope of the disclosure. Thus, the present disclosure is not intended to be limited to the aspects shown herein but is to be accorded the widest scope possible consistent with the principles and novel features as defined by the following claims.

What is claimed is:

1. A host device comprising:
system memory including a logical-to-physical (L2P)
cache and a second cache; and
a host controller interface (HCI) configured to be coupled
to a flash memory device and configured to:
copy a particular region of a L2P address mapping table
to the L2P cache, the L2P address mapping table
configured to include mappings between logical
memory addresses and physical memory addresses
of the flash memory device;
determine that the particular region is to be removed
from the L2P cache;
in response to the determination that the particular
region is to be removed, identify a particular sub-
region to be retained that has an access metric that
satisfies a retention criterion, wherein the particular
sub-region is a sub-region of the particular region;
in response to the identification of the particular sub-
region, store the particular sub-region into the second
cache; and
after the storage of the particular sub-region into the
second cache and in response to the determination
that the particular region is to be removed, remove
the particular region from the L2P cache.
2. The host device of claim 1, wherein the L2P cache
corresponds to a Host Performance Booster (HPB) cache,
and wherein the second cache is implemented in an inacti-
vated region partition of the system memory.
3. The host device of claim 1, wherein the HCI is
configured to, in response to a cache miss resulting from a
first cache lookup at the L2P cache, perform a second cache
lookup at the second cache.
4. The host device of claim 1, wherein the HCI is
configured to:
after the storage of the particular sub-region into the
second cache, perform a cache lookup at the second
cache including comparing a logical address of data to
be accessed to logical addresses corresponding to sub-
regions stored in the second cache; and
in response to detecting a match of the logical address to
particular logical addresses of the particular sub-region
stored in the second cache, read a physical address
associated with the logical address from the particular

32

sub-region and send the physical address to the flash
memory device in conjunction with a request to access
the data.

5. The host device of claim 1, wherein the HCI is
configured to remove a second region from the L2P cache in
response to a notification from the flash memory device that
the second region is to be removed.

6. The host device of claim 1, wherein the HCI is
configured to determine that the particular region is to be
removed in response to a cache miss resulting from a cache
lookup of a logical address at the L2P cache and at the
second cache.

7. The host device of claim 1, wherein the access metric
is based on a count of accesses to the particular sub-region.

8. The host device of claim 1, wherein the HCI is further
configured to maintain a counter for each sub-region of the
particular region, wherein the counter for a sub-region is
updated each time that sub-region is accessed.

9. The host device of claim 1, wherein the HCI is
configured to identify the particular sub-region responsive to
determining that a counter associated with the particular
sub-region has a highest counter value of counters that are
associated with sub-regions of the particular region.

10. The host device of claim 1, wherein the HCI is
configured to determine that the retention criterion is satis-
fied based on determining that the access metric exceeds an
access count threshold.

11. The host device of claim 1, wherein the access metric
is stored in the L2P cache.

12. The host device of claim 1, wherein the HCI is
configured to:

responsive to determining that the second cache is full,
determine that a second sub-region is to be removed
from the second cache; and
remove the second sub-region from the second cache,
wherein the particular sub-region is stored into the
second cache after removing the second sub-region
from the second cache.

13. The host device of claim 6, wherein the HCI is
configured to, in response to the cache miss:
send the logical address to the flash memory device;
receive a replacement region that includes a first sub-
region associated with the logical address; and
after removing the particular region from the L2P cache,
store the replacement region in the L2P cache.

14. The host device of claim 12, wherein the HCI is
configured to determine that the second sub-region is to be
removed responsive to determining that the second sub-
region has a second access metric that satisfies a sub-region
eviction criterion.

15. The host device of claim 14, wherein the second
access metric includes a first sub-region added to the second
cache, a less than threshold access count, or both.

16. A method comprising:

at a host device coupled to a flash memory device:
copying a particular region of a logical-to-physical
(L2P) address mapping table to a L2P cache of the
host device, the L2P address mapping table config-
ured to include mappings between logical memory
addresses and physical memory addresses of the
flash memory device;
determining that the particular region is to be removed
from the L2P cache;
in response to determining that the particular region is
to be removed, identifying a particular sub-region to
be retained that has an access metric that satisfies a

33

retention criterion, wherein the particular sub-region is a sub-region of the particular region;
 in response to identifying the particular sub-region, storing the particular sub-region into a second cache of the host device; and
 after storing the particular sub-region in the second cache and in response to determining that the particular region is to be removed, removing the particular region from the L2P cache.

17. The method of claim 16, wherein the L2P cache corresponds to a Host Performance Booster (HPB) cache in a system memory of the host device, and wherein the second cache is implemented in an inactivated region partition of the system memory of the host device.

18. The method of claim 16, further comprising, in response to a cache miss resulting from a first cache lookup at the L2P cache, performing a second cache lookup at the second cache.

19. The method of claim 16, further comprising performing a cache lookup at the second cache including:

after storing the particular sub-region into the second cache, comparing a logical address of data to be accessed to logical addresses corresponding to sub-regions stored in the second cache; and

in response to detecting a match of the logical address to the particular sub-region stored in the second cache, reading a physical address associated with the logical address from the particular sub-region and sending the physical address to the flash memory device in conjunction with a request to access the data.

20. The method of claim 16, further comprising removing a second region from the L2P cache responsive to a notification from the flash memory device that the second region is to be removed.

21. The method of claim 16, wherein the particular region is to be removed is determined responsive to a cache miss resulting from a cache lookup of a logical address at the L2P cache and at the second cache, and further comprising, in response to the cache miss:

sending the logical address to the flash memory device; receiving a replacement region that includes a first sub-region associated with the logical address; and after removing the particular region from the L2P cache, storing the replacement region in the L2P cache.

22. The method of claim 16, wherein the access metric is based on a count of accesses to the particular sub-region.

23. The method of claim 16, further comprising maintaining a counter for each sub-region of the particular region, wherein the counter for a sub-region is updated each time that sub-region is accessed.

24. The method of claim 16, wherein the particular sub-region is identified responsive to determining that a counter associated with the particular sub-region has a highest counter value of counters that are associated with sub-regions of the particular region.

25. The method of claim 16, wherein satisfying the retention criterion includes determining that the access metric exceeds an access count threshold.

26. The method of claim 16, wherein the access metric is stored in the L2P cache.

34

27. The method of claim 16, further comprising:
 responsive to determining that the second cache is full, determining that a second sub-region is to be removed from the second cache; and

removing the second sub-region from the second cache, wherein the particular sub-region is stored into the second cache after removing the second sub-region from the second cache.

28. The method of claim 27, wherein the second sub-region is to be removed is determined responsive to determining that the second sub-region has a second access metric that satisfies a sub-region eviction criterion, and wherein the second access metric includes a first sub-region added to the second cache, a less than threshold access count, or both.

29. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to:

copy a particular region of a logical-to-physical (L2P) address mapping table to a L2P cache of a host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device; determine that the particular region is to be removed from the L2P cache;

in response to the determination that the particular region is to be removed, identify a particular sub-region to be retained that has an access metric that satisfies a retention criterion, wherein the particular sub-region is a sub-region of the particular region;

in response to the identification of the particular sub-region, store the particular sub-region into a second cache of the host device; and

after the storage of the particular sub-region into the second cache and in response to the determination that the particular region is to be removed, remove the particular region from the L2P cache.

30. An apparatus comprising:

means for copying a particular region of a logical-to-physical (L2P) address mapping table to a L2P cache of a host device, the L2P address mapping table configured to include mappings between logical memory addresses and physical memory addresses of a flash memory device;

means for determining that the particular region is to be removed from the L2P cache;

means for identifying, in response to a determination that the particular region is to be removed, a particular sub-region to be retained that has an access metric that satisfies a retention criterion, wherein the particular sub-region is a sub-region of the particular region;

means for storing, in response to an identification of the particular sub-region, the particular sub-region into a second cache of the host device; and

means for removing, after storage of the particular sub-region into the second cache and in response to the determination that the particular region is to be removed, the particular region from the L2P cache.

* * * * *